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**Driving a Pair of Worn Robot Arms: Interface Design,
Safety, and What the Machine Can and Cannot Reach**

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Abstract

This project is about a pair of robot arms that a person wears on their back, and about the problem of letting somebody else drive them safely.

The arms are two seven-jointed Kinova Gen3 manipulators on a backpack frame. One person wears them. A second person, standing across the room, drives them. That split is deliberate, and it changes the problem: the wearer cannot see what the driver intends, cannot feel the controls, and did not choose the motion happening beside their head. Everything in this thesis follows from treating the wearer as a second participant rather than as part of the apparatus.

Three ways of driving the arms were built and compared. The first is a mannequin arm fitted with sensors, which the driver moves by hand. The second uses a pair of hand controllers borrowed from a virtual reality headset, held as motion trackers rather than worn. The third takes a sentence typed or spoken in ordinary English and works out the rest. Around all three sits a safety layer whose job is to keep the metal a fixed distance away from the person carrying it.

Three findings shaped the outcome. The first is that the mannequin arm loses sensor channels as its wiring flexes, so the software was built to keep working with fewer of them, to say which ones it no longer trusts, and to refuse rather than guess when too few remain. Seven of fourteen channels currently fail the health test, and replaying the failing ones with the driver's hand held still shows them injecting 54.5 mm of movement that nobody asked for.

The second is a limit of the machine rather than of the software. With the arms mounted where they are, there is no point in front of the wearer's chest that both arms can reach, and the innermost useful working column sits 325 mm to one side of the body centre line. That result was reached four separate ways, including with the wearer's own arms deleted from the model, and it rules out any task that needs the two hands to meet. A mount change that would improve it is measured and costed, and deliberately not applied.

The third concerns measurement itself. Over six weeks, a surprising result turned out to be a fault in the measuring tool rather than in the robot on seventeen separate occasions. The habit that came out of this, of clearing the instrument before believing a bad number, is described in its own chapter with the cases that produced it.

No result here has been validated on a moving physical arm carrying a load, and no human data has been collected. The study protocol, the tasks, the measures and the analysis are complete and are reported in full; [Chapter 9](#) is deliberately left empty, with its tables and figures in place, so it can be filled the moment ethical approval and a working master arm allow the sessions to run.

Acknowledgements

I thank my supervisor, Dr. Dandan Zhang, for steering this project toward what could be measured rather than what could be asserted. Several of the results here exist only because a number was questioned before it was written down.

I thank my co-supervisor, Dr. Etienne Burdet, for the reframing that gave the project its shape. Treating the wearer and the driver as two people with separate interests, rather than as one role, is the reason the safety and workspace chapters look the way they do.

I am grateful to the Human Robotics Group for access to the platform, and to everyone who kept two arms and a backpack frame running.

To be completed before submission: name the technical staff, laboratory members and fellow students who gave practical help, and anyone who lent equipment or time on the platform. Only the two supervisors and the group are named above because those are the contributions on record.

A note on the words used

This thesis is written to be read by somebody who does not work on robots. A small number of technical words cannot be avoided without making the sentences longer than the ideas in them, so they are defined here once and then used plainly.

Table 1: The technical vocabulary used in this thesis, in the sense meant here.

Word	What it means here
Wearer	The person carrying the arms on their back. A participant, not equipment.
Driver, or operator	The person commanding the arms. A different person from the wearer.
Master arm	The mannequin arm fitted with sensors that the driver moves by hand. The robot copies it.
Joint	One place where the arm bends or twists. Each robot arm has seven, plus a gripper.
End effector	The point between the gripper's two finger pads, which is where the arm actually touches things.
Pose	A position and an orientation together. Where the hand is, and which way it is pointing.
Inverse kinematics	Working out the seven joint angles that put the hand at a wanted pose. There is usually more than one answer.
Reachable	A pose the arm can adopt without hitting anything and without running out of joint travel.
Clearance	The shortest distance from any moving part of the arm to any part of the wearer's body.
Clearance floor	The smallest clearance the system will allow, set to 150 mm.
Simulation	A model of the arms and the wearer running on a computer, used to test motion before any metal moves.
Node	One running program in the software. The system is built from about forty of them, each doing one job.
Trial	One attempt at one task by one pair of people, under one way of driving.
Dyad	One wearer and one driver, working together. The unit of analysis in the study.

Two conventions are used throughout. Where a number is quoted, the program that produced it is named, so it can be recomputed. Where a quantity has never been measured, the text says so rather than offering an estimate, because a figure with a guess inside it hides the thing that ought to be measured next.

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Chapter 1

Introduction

An extra robot arm worn on the body is not a prosthesis. The person keeps the limbs they were born with, and the machine works alongside them [26, 23]. The idea is old enough to have a literature and young enough that most of it is still about building the hardware rather than living with it.

What makes these devices awkward to control is easy to state. A robot arm bolted to a bench has a base that stays where it was put. A robot arm bolted to a person has a base that breathes, sways, turns to look at something and occasionally flinches. It is also soft, irreplaceable, and within arm's reach of the thing being moved.

This project works on the platform at Imperial College London known as Dr. Octopus [10], which carries two Kinova Gen3 seven-jointed arms [12] on a backpack frame. The arms are worn by one person and driven by another. That arrangement is the starting point rather than a detail, and Fig. 1.1 shows how the parts sit together.

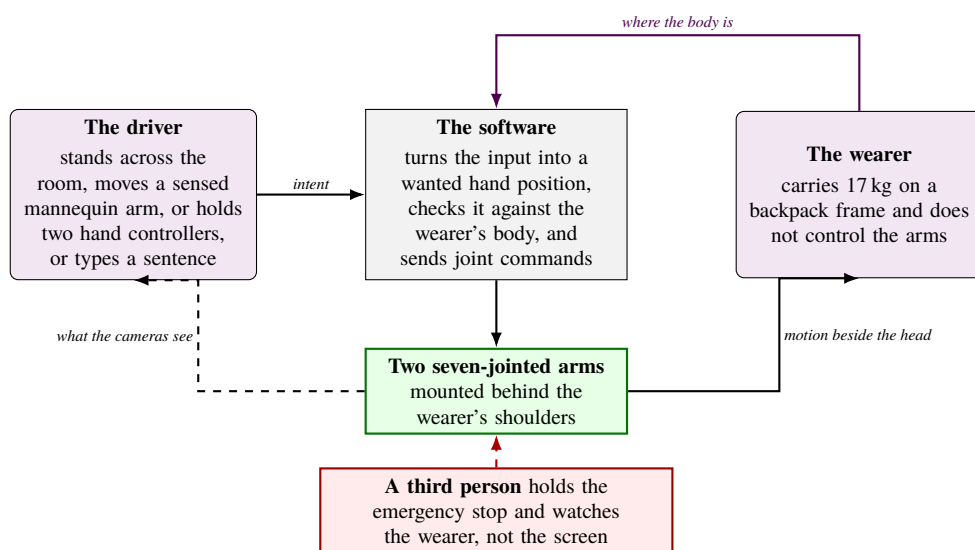


Figure 1.1: The arrangement this thesis is about. Two people, not one. The driver commands the arms and cannot feel them; the wearer carries them and does not command them. The software between the two is responsible for the wearer's safety, because neither person is in a position to guarantee it.

1.1 The problem

Two arms of seven joints each give fourteen things that can move. A person has two hands and a limited amount of attention. Closing that gap is the design problem, and there is no way to do it that does not give something up.

The obvious approach is to give the driver a small copy of the robot and let them pose it. That is the approach taken here, and it works up to a point. The mannequin master reproduces the sequence of joints in the real arm at roughly a third of the size. What it cannot reproduce is the accuracy: it routes fourteen analogue signals through joints that turn, and wires that flex through turning joints eventually stop reporting. By the time this project was measuring rather than building, seven of the fourteen channels no longer produced a usable signal.

That could be treated as a repair job. It was instead treated as the normal operating condition, because the same thing will happen again to whichever channels are working next month. The system therefore has to answer three questions at all times: which channels can be believed, what the arm can still be commanded to do with the ones that remain, and what it should do when the answer is nothing.

The second problem is the wearer. Standard robot software checks whether a planned motion would touch anything, and reports either contact or no contact. A person standing inside a pair of moving arms needs more than that. They need a margin, held all the way along the path and not only at the ends, and they need it enforced against the parts of the arm that actually swing past them, which are the ones nearest the shoulder rather than the hand. Getting this right turned out to be harder than expected, and [Chapter 5](#) is largely an account of the ways it can be got wrong while appearing to work.

1.2 Aims

1. Build a sensed mannequin arm that reproduces the robot's shape at reduced size, and characterise what its sensors actually deliver rather than what they are specified to deliver.
2. Turn the driver's hand movement into arm commands in a way that keeps working as sensor channels fail, and measure what each failure costs.
3. Build a safety layer suited to a machine mounted beside a person, and test it by deliberately breaking things rather than by reading the code.
4. Measure where the arms can actually work once the wearer's body is taken into account, and say plainly which tasks that rules out.
5. Add a second and third way of driving the arms, so that the study has something to compare against, and check that each reaches the same commands.
6. Design the two-person study in full, and identify by measurement what has to be fixed before it can be run.

1.3 What is new here

Three things in this thesis are not in the work it builds on.

The first is the treatment of sensor loss as a designed-for condition. Systems of this kind are usually described in their working state. Here each channel carries a health verdict computed from its own recent behaviour, the checks applied to an incoming frame depend on which channels the current mode actually reads, any channel can be switched off while running, and the cost of switching it off is published rather than assumed. When too little is left, the system reports that it has no position rather than inventing one. On a machine beside somebody's head, an invented position is worse than an admitted gap.

The second is a negative result with a price attached. Under the mount as built, the two arms share no reachable point anywhere in front of the wearer. That is not a tuning failure and no amount of better control will change it.

It is reported here with the four independent measurements that establish it and with a costed mechanical change that would improve it.

The third is methodological, and it is the part most likely to be useful elsewhere. Seventeen times during this project, a result that looked like a fault in the robot turned out to be a fault in the thing measuring the robot. [Chapter 8](#) reports the numbers; the habit that produced them, and the cases that taught it, are set out in [Section 8.1](#).

1.4 How this thesis is arranged

[Chapter 2](#) covers the work this builds on and states what is left open. [Chapter 3](#) describes the system as built, in overview. [Chapter 4](#) covers the mannequin arm and how a hand movement becomes an arm command. [Chapter 5](#) covers the safety layer. [Chapter 6](#) covers the cameras, and the change from a robot that was told where things are to one that measures where they are. [Chapter 7](#) sets out the study: tasks, conditions, measures and ethics. [Chapter 8](#) reports everything measured on the system itself. [Chapter 9](#) is where the results from people will go, and is empty. [Chapter 10](#) weighs it up, and [Chapter 11](#) says what should happen next and in what order.

Chapter 2

Background

Five bodies of work bear on this system: worn robot limbs and what they are for, the split between the person wearing a machine and the person driving it, the problem of driving a robot from a control that is not the same shape, sharing control between a person and a program, and how the wearer's experience is measured. This chapter takes them in that order and finishes with what is left open, which is narrower than it first looks.

2.1 Robot limbs worn on the body

The field's founding demonstrations come from Parietti and Asada, whose extra limbs brace against the environment to steady the wearer while they work overhead [19, 18], later extended to supporting the whole body [20]. Bracing is a useful starting point because it names the mechanical difficulty at once. An arm mounted on a person pushes back into that person, and the person moves.

Later reviews map the design space along consistent lines [26, 23, 25, 27]. Where the limb is mounted trades reach against the twisting load it puts on the wearer. How it is actuated trades responsiveness against weight. How it is commanded ranges from muscle signals to a deliberate manual control. A recent survey adds a pointed observation: the field reports how embodied the wearer feels far more often than it reports whether the task got done [30].

Weight binds harder than anything else and is discussed least. Each Gen3 arm is 8.2 kg, so the pair with the harness passes 17 kg. Simulation of the loads such systems place on the body suggests this is not a detail [17], and it is the main reason the session design in [Chapter 7](#) caps how long anybody wears the rig.

SPACE RESERVED: FIGURE FROM PUBLISHED WORK

Parietti and Asada's bracing arrangement, with the limbs anchored against the work surface so that the wearer, the limbs and the environment form one closed chain. Source: [19].

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Figure 2.1: The arrangement that set the mechanical framing for the field. To be reproduced from [19] once permission is obtained.

2.2 When the wearer and the driver are different people

The closest predecessor to the arrangement used here is Fusion, which put a robot head and two arms on a worn frame and let a remote operator drive them [24]. Two qualifications have to travel with that reference. Its three modes describe how firmly the robot arms are strapped to the wearer’s own arms, not how much the machine decides; in all three the machine decides nothing. And it is a two-page demonstration with no user study and no performance figures. It shows the arrangement existed. It cannot support a claim about how well it worked.

The evidence about how it feels to wear one of these comes from elsewhere. Recent work on how close such a limb should come to the body ran a hidden human operator, varied how autonomous the machine appeared to be, and measured the wearer’s arousal and reported trust [31]. The result runs against intuition. Greater apparent autonomy made both worse: arousal up, trust down. Because the autonomy in that study was a concealed person and the wearer had nothing else to do, it does not settle what happens when the autonomy is real and the wearer is busy. It does establish that the wearer’s experience cannot be assumed benign.

Part of the two-person problem is already solved. Zhang et al. compensate for the motion a human base introduces and report end-effector errors around 1.4 mm with the shoulder as a moving base [28]. What that leaves open is the case here, where the disturbance comes from a nervous system the driver has no advance warning of. When one person both wears and drives, the sway is partly predictable to the controller because the same person caused it. With two people it is not.

2.3 Driving something that is not the same shape

If the control and the robot differ in size or proportion, the mapping between them has to be chosen. Guggenheim and Asada argue for small physical copies of the robot on the grounds that a similar shape gives the operator feedback for free, without a separate force display [8]. The argument holds for similarity of shape and says nothing about similarity of size, which is where this system departs: the master’s links total 0.272 m against the robot’s 0.902 m of reach, so the same joint angles produce very different hand movements.

Two mappings are available. Sending master joint angles straight to robot joint angles preserves the shape the arm takes, which matters when the shape itself has to avoid something, but it demands that every master joint be measured accurately. Mapping hand movement to hand movement decouples the two chains and leaves the robot’s spare joint unspecified. [Chapter 4](#) records why this project started with the first and ended with the second.

A related question is what to feed back to the driver. Pinardi et al. review the added sensory channels used in movement augmentation and find that improving control and improving the sense of ownership do not go together [21]. Neither is a proxy for the other, which is why [Chapter 7](#) measures both.

2.4 Sharing control with the machine

Shared autonomy asks a different question from teleoperation: not how to transmit the operator’s intent faithfully, but how much of it the machine should supply. Policy blending mixes the operator’s command with the machine’s estimate of the goal in proportion to confidence [6]. Hindsight optimisation treats it as a decision under uncertainty about which goal is meant [11]. Both assume a bench robot and an operator who can see it directly.

Two things change when the robot is worn by a second person. The cost of guessing wrong is no longer a dropped object, and the operator’s confidence is no longer the only confidence that matters. The design used here keeps position entirely under the driver’s control in every mode and lets the machine supply only the wrist

orientation, which is the part the failing sensor channels cannot deliver anyway. That choice is argued in [Chapter 3](#) and is the reason the assistance can be switched on without changing who is responsible for where the hand goes.

2.5 Measuring the wearer's experience

Three instruments recur in this literature and are used unchanged here. Task load is measured with the NASA Task Load Index [9], physical exertion with the Borg category-ratio scale [3], and the sense that a limb belongs to you with the embodiment questionnaire developed from the rubber-hand work [14, 4]. Difficulty of a pointing task is indexed in the standard way [7], which lets tasks of different sizes and distances be compared.

2.6 What is open

It is tempting to claim that nobody has looked at a two-person worn limb with real autonomy. That would be too strong. The concealed-operator study [31] has already varied autonomy and measured the wearer, and Fusion [24] already built the two-person arrangement. What has not been done is to measure task performance and wearer experience together, in the same sessions, with autonomy that is genuinely a program and a wearer who has a job of their own. That is the gap this study is designed to fill, and [Chapter 7](#) sets out how.

Chapter 3

The system as built

This chapter describes what exists, at the level of what each part is for. The two chapters after it go into the parts that carry the most risk.

3.1 The hardware

Two Kinova Gen3 arms, each with seven joints, a two-finger gripper and a small colour-and-depth camera at the wrist, are carried on a backpack frame. The mounting brackets sit behind the wearer's shoulders and tilt the arms forward so they come over the top rather than out to the sides.

The driver's mannequin arm is a scale model of the same joint sequence with a rotary sensor at each joint and a small inertial sensor at the wrist. A Teensy 4.1 microcontroller [22] reads all of it and streams the numbers over a USB cable at about 50 Hz. There is no force feedback of any kind: the driver feels the mannequin arm's own friction and nothing about what the robot is touching.

Everything runs on one laptop under Linux, with the robot software built on ROS 2 [16]. That machine has a small graphics card with 4 GB of memory, which decided several questions later on by ruling out the larger vision and planning packages outright.

SPACE RESERVED FOR A PHOTOGRAPH

The platform as assembled: the backpack frame worn on a stand, both arms mounted and folded to the presentation pose, the mannequin master on the bench beside it with its cable run, and the emergency stop within reach. One wide shot, and one close view of a shoulder bracket showing the 60° forward tilt.

Figure 3.1: The platform. A photograph is needed here rather than a rendering, because the mounting bracket and the cable routing are the two things a reader cannot infer from the model.

3.2 The software, in one picture

The software is about forty small programs that talk to each other over named channels. They are grouped into eight packages by what they are responsible for, and the grouping matters for one reason: the teleoperation package is not allowed to depend on any of the others.

That rule sounds like tidiness and is not. Plain teleoperation is the baseline that every assisted condition in the study is compared against. If the teleoperation code called into the autonomy code, the comparison would be between the autonomy and itself.

Table 3.1: The eight first-party packages and what each is responsible for. Sizes are measured lines of Python.

Package	Responsible for	Lines
Teleoperation	Reading the master arm, turning it into commands, the clutch, motion smoothing, the emergency stop, the operating window	25 737
Perception	Cameras, finding objects, measuring the table, building a map of the scene, tracking the wearer	9 431
Autonomy	Choosing a grasp, guessing the goal, handing control over, understanding a sentence	3 886
Experiments	Tasks, conditions, logging, sessions, analysis	12 278
Description	The model of the arms, the frame and the wearer	—
Motion planning	Inverse kinematics, controllers, collision settings	—
Controller bridge	The hand controllers and their safety checks	2 709
Controller autonomy	Assistance on the controller path	246

3.3 From a hand movement to a moving arm

[Figure 3.2](#) is the path every command takes, whichever way the arms are being driven. It is worth reading once carefully, because most of the rest of this thesis is about one or another of its boxes.

The path is deliberately short. The wanted hand pose goes to a solver that returns joint angles, the whole arm is checked against the wearer, the result is smoothed so no joint is asked to move faster than it can, and the joint commands go out. There is no planner in that loop and there should not be: the driver's hand is the plan, and the right problem is to follow it smoothly. For the autonomous mode, where there is a goal to plan towards, a planner was added and is described in [Chapter 6](#).

3.4 Three ways to drive

The mannequin master is the baseline. The driver moves it and the robot copies the movement of its tip, scaled and offset.

The hand controllers came from a virtual reality headset, but the way they are used has almost nothing to do with virtual reality. Nobody wears the headset. It stands on a shelf and acts as the tracking reference while the driver holds the two controllers and watches the real robot across the room. That arrangement is the one the laboratory actually uses, and it caused two specific safety problems that are dealt with in [Section 5.6](#).

The third route takes a sentence. Typing *put the blue block on the blue mat* produces a plan, shows the driver what it understood before anything moves, and waits for a confirming click. It is built on a grammar rather than a language model, for three reasons: a grammar can explain why it refused, it can be tested exhaustively against

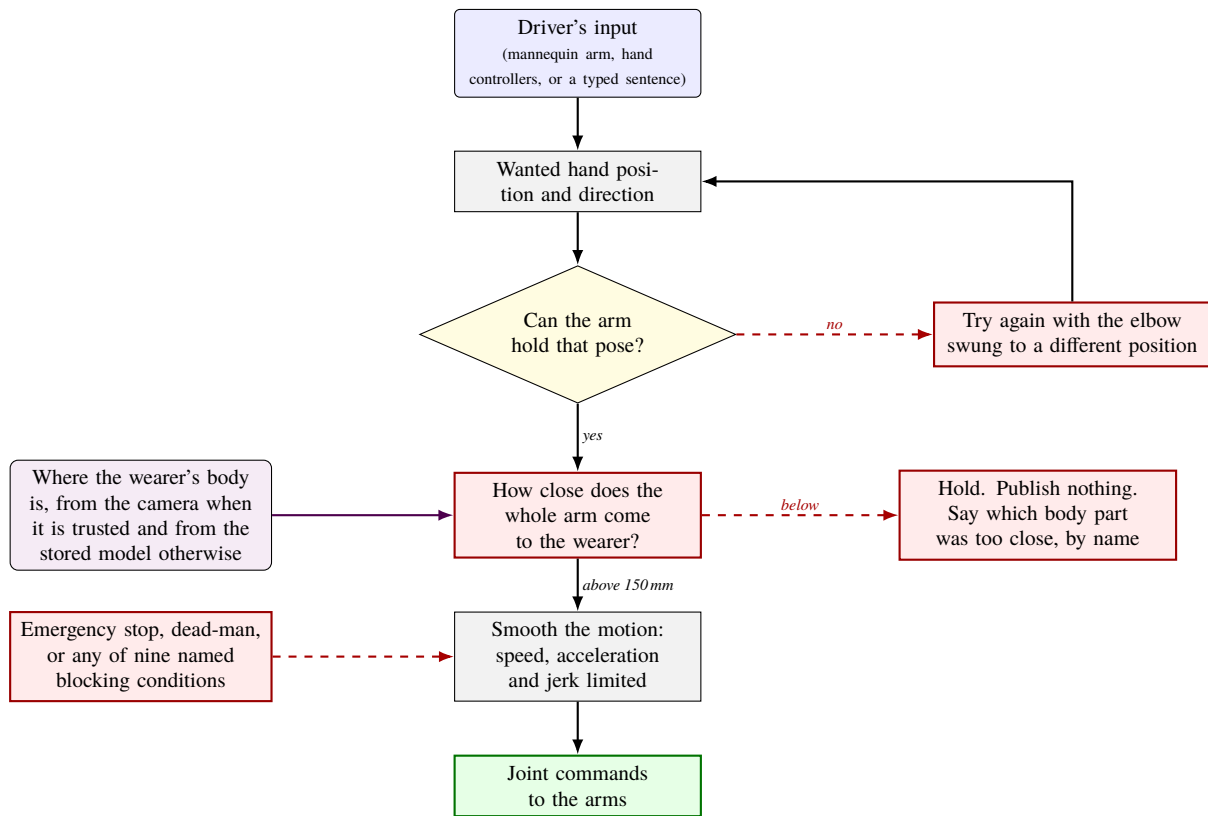
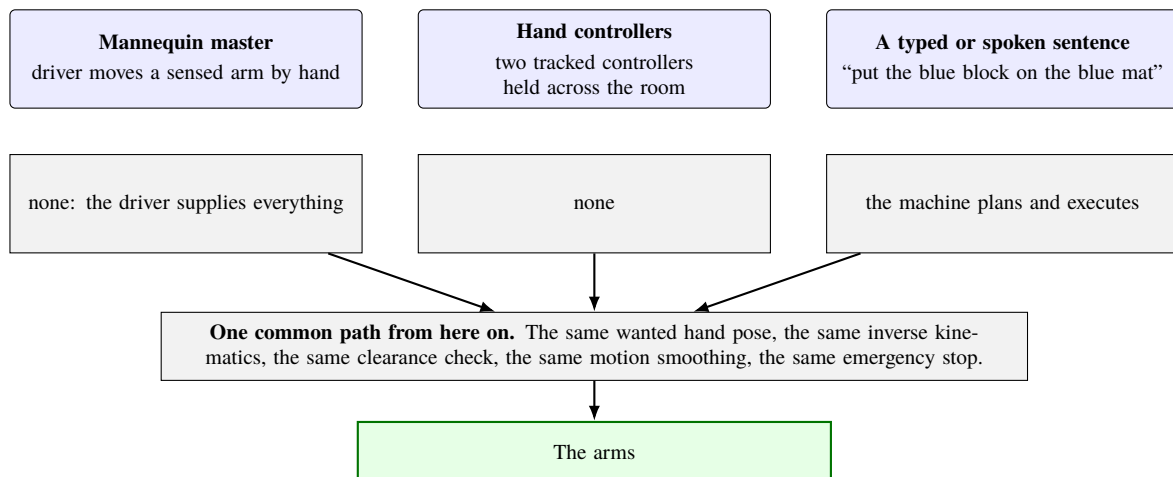


Figure 3.2: The path from the driver's input to the arms. Red boxes can stop the motion; the green box is the only one that moves it. Two of the red boxes matter most: the retry that swings the elbow to a different position when the first attempt fails, and the clearance check, which is measured against the wearer's body geometrically rather than being taken from the collision settings.



Assistance is added only above the common path, never inside it. Whichever way the arms are being driven, the part that keeps the wearer safe is the same code running the same checks.

Figure 3.3: The three input routes. They differ only above the common path. Everything that protects the wearer is shared, so a change to the safety layer cannot apply to one condition of the study and not another.

a list of phrasings with a scored outcome for each, and it fits in the memory the graphics card has left after the object detector has loaded.

3.5 The operating window

Everything reachable from a terminal is also reachable from one window, and that is a rule rather than a convenience. A feature that can only be started by typing a command does not exist for the person running a session, who has a wearer standing in a rig and no attention to spare.

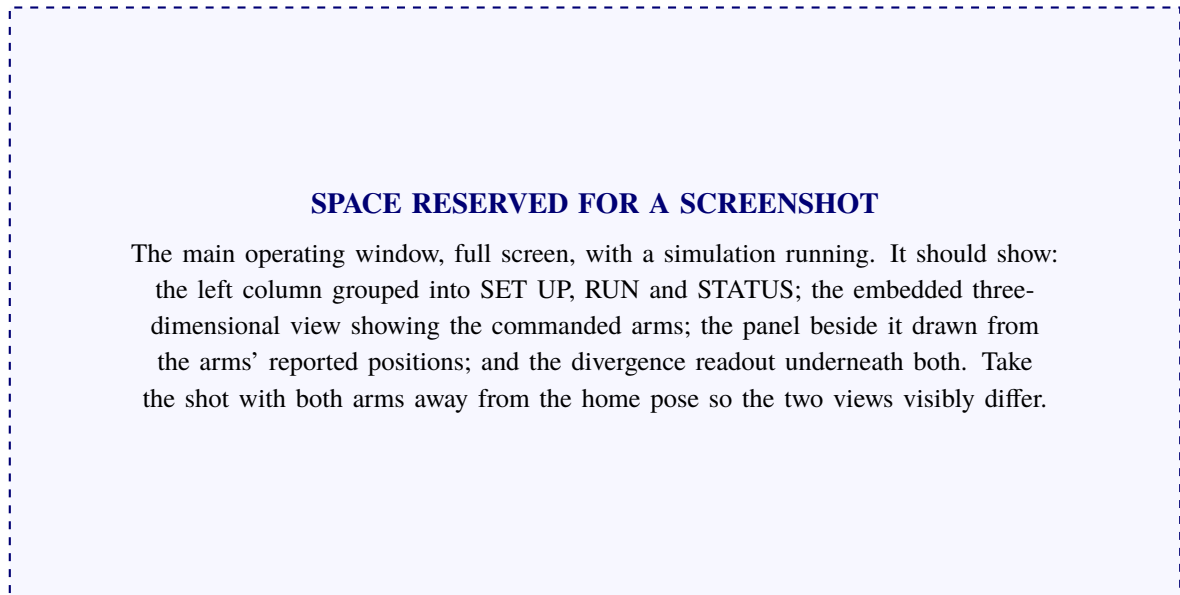


Figure 3.4: The window used to run a session. The point of the side-by-side views is that the left one shows what the arms were told to do and the right shows what they report doing, so a difference is visible without reading a log.

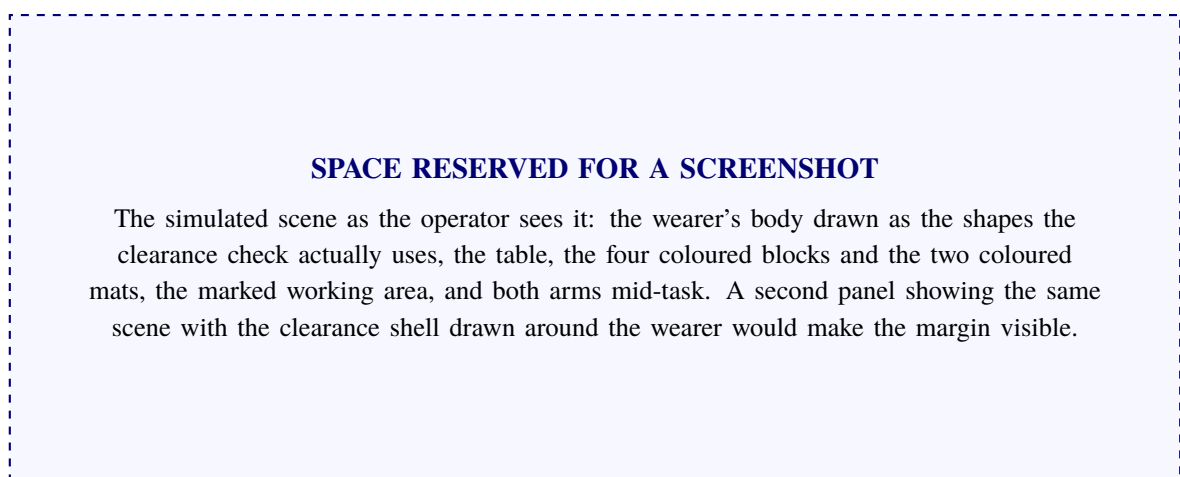


Figure 3.5: The simulation. The wearer is drawn from the same description the safety check measures against, so what is on screen and what is being enforced cannot drift apart.

The window is checked by a program that presses every control in it and then checks what the press did, currently 257 separate checks. That sounds excessive until one notices that the first time it ran end to end it found

four faults, including a set of buttons that had been doing nothing at all because the error they raised was being swallowed silently.

Chapter 4

The mannequin arm and how it commands the robot

4.1 What the master arm is

The master is a scale model of the robot arm, about a third of the size, with the same sequence of joints. A rotary sensor sits at each joint and a small inertial sensor sits at the wrist, and a microcontroller reads them all and sends the numbers up a USB cable. The driver holds it by the tip and moves it. Nothing pushes back.

4.2 Why the joints are not copied directly

The first version of this system sent each master joint angle straight to the matching robot joint. That is the simplest thing to build and it was abandoned, for a reason that is easy to see once stated: the two arms are not the same size. A 10° bend at the master's elbow moves its tip a few centimetres and the robot's hand about three times as far. Worse, the robot's joint limits and the master's are different, so there are master poses with no robot equivalent at all.

What replaced it maps movement to movement. The driver's hand moving 100 mm forward asks the robot's hand to move 100 mm forward, from wherever it happens to be. The robot's own solver works out the joint angles. This decouples the two arms completely and costs one thing: the robot's spare joint, the one that swings the elbow without moving the hand, is left unspecified and has to be chosen by something else.

4.2.1 Lifting the mouse

Because the master's reach is a third of the robot's, the driver runs out of arm before the robot runs out of workspace. The fix is the one a computer mouse uses. A button disengages the link, the driver brings their hand back to a comfortable position, and pressing again re-engages from there. The robot does not move during the disengagement.

Getting this right took three separate corrections. The button had to be debounced, because a single press was registering twice. The reference position had to be captured only when the master was genuinely still, otherwise the system latched onto a moving target. And the smoothing filter had to be cleared on re-engagement, or the first few frames after re-engaging were a blend of the new position and the old one. Measured over eight re-engagements, the arm now travels the full distance each time with no snap-back.

Master arm chain — roll / bend alternation
lengths in mm, CAD-inferred against measured

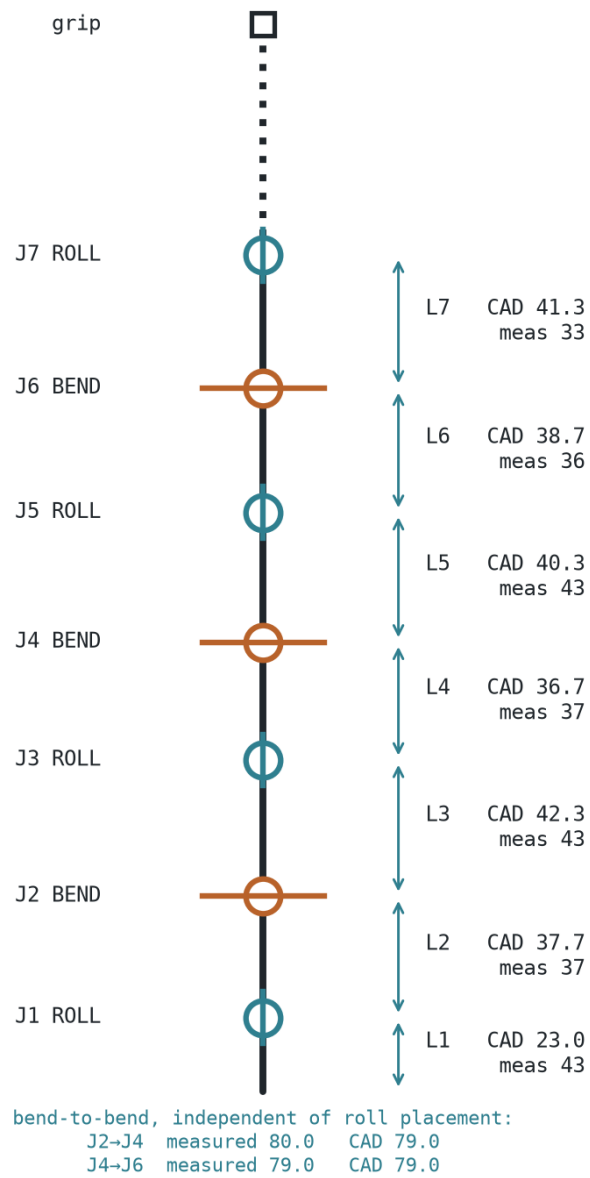


Figure 4.1: The master arm as modelled, with its joint sequence labelled. Its links total 0.272 m against the robot's 0.902 m of reach, so the same joint angles do not produce the same hand movement at each end.

4.3 Reading the sensors: what the channels actually do

The specification says fourteen rotary sensors. What arrives is less tidy.

A failing sensor of this kind does not go quiet. It reports values that span the full range and are not a trajectory: consecutive readings unrelated to each other, jumping tens of degrees in a fraction of a second. Judged on range alone, such a channel looks healthy, which is exactly how a broken one keeps being trusted. The system therefore classifies each channel into one of four states, and the third of them had to be invented for this project.

Table 4.1: The four verdicts, and the current state of the master arm. Measured over 20 430 recorded frames.

Verdict	Meaning	Channels
Working	Reports a smooth range of values, rarely drops out	5 of 14
Dropping out	Reports usable values with gaps	1 of 14
Incoherent	Present, fresh, and wrong: more than one update in twenty jumps by over 60°	6 of 14
Dead	No usable reading at all	2 of 14

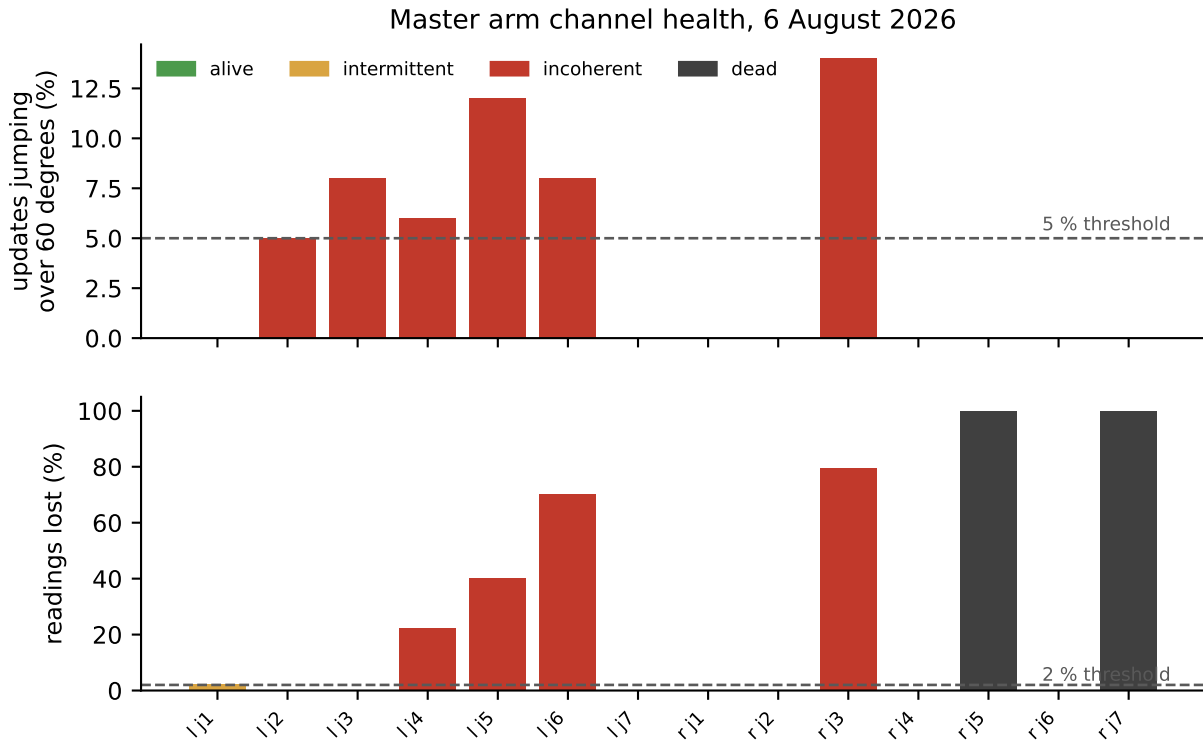
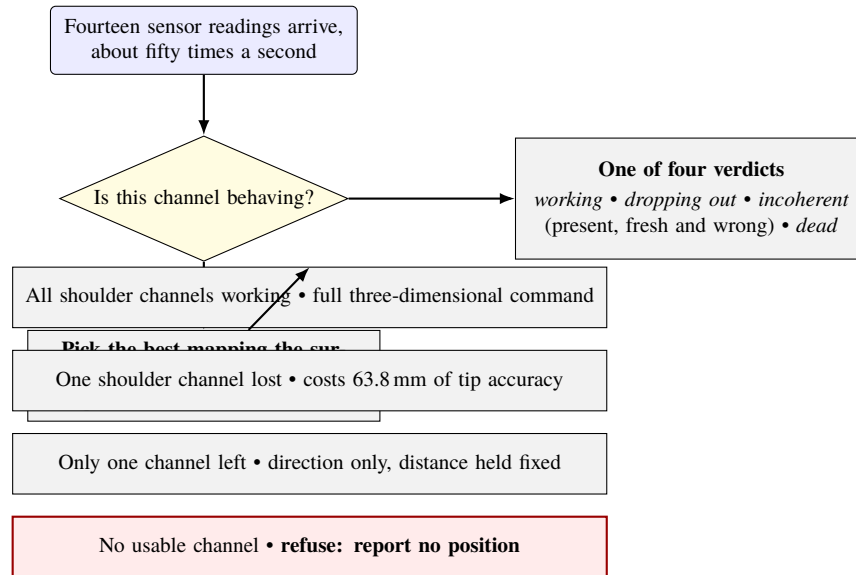


Figure 4.2: Every channel, on both measures that decide its verdict. The upper panel is the one that matters: a channel above the five per cent line is reporting values that are present, fresh and not a trajectory. Six channels sit above it, five of them on the left arm. Produced from the stored channel record of 6 August 2026.

One correction belongs here because it changed two conclusions that had stood for weeks. The recorder writes rows at about 50 Hz while the sensors themselves update at fifteen to seventeen. Between four and five of every six rows are therefore an exact copy of the row before. Any statistic computed across consecutive rows is dominated by those copies, which is why the noise figure read as exactly zero on every channel. It was measuring the recorder. Computed between genuine sensor updates instead, the noise floor is 0.2 to 5.4°, and one channel that had been written off as dead turned out to be the cleanest one on that arm.

4.4 Working with fewer channels

Figure 4.3 shows what happens as channels are lost. The system does not have a working state and a broken state. It has a ladder, and it announces which rung it is standing on.



The refusal at the bottom is the important rung. On a machine mounted beside somebody's head, a position that has been guessed is worse than an admitted gap, and the difference has to be visible from outside the program.

Figure 4.3: What the system does as sensor channels fail. Each rung is a different way of estimating where the driver's hand is, using fewer inputs than the one above it. The cost of each step down is measured rather than assumed, and the bottom rung is a refusal.

The mapping used at the top of the ladder splits the driver's hand position into three parts and takes each from whichever sensor sees it most directly. How high the hand is comes from the wrist's inertial sensor, which feels gravity and therefore needs no calibration and no drift correction. Which way round the hand is comes from the shoulder joint, which is a single rotary sensor. How far out the hand is comes from the remaining joint angles. The point of splitting it this way is that losing one part does not destroy the others.

Two numbers say what this costs. Dropping from four sensed joints to one costs 63.8 mm of accuracy at the tip, measured against the recorded frames. And replaying the failing channels with the driver's hand deliberately held still shows them injecting 54.5 mm of commanded movement that nobody asked for, against a master reach of 272 mm. The second figure is the reason the failing channels are frozen rather than filtered: the arm is a relay, and whatever the sensors feed in, the robot faithfully reproduces.

4.5 Making the motion smooth

Until late in the project, the last step before the joint commands went out was a simple limit on how far each joint could move in one cycle. It is the obvious approach and it has a defect that is not obvious.

Because each joint is limited separately, a joint with a short distance to travel arrives in one cycle while a joint with a long distance takes two. In between, the joints are at different fractions of their own journeys, and the hand is not on the straight line the solver intended. Measured on a typical move, the hand strayed 51.3 mm from the line it was supposed to follow. For comparison, the gripper has to be within 30 mm of an object to close on it.

The fix was to replace the per-joint limit with a generator that plans all seven joints together, so they start together, finish together, and stay in proportion in between [2]. The straying went to zero, every joint now arrives on the same cycle, and the peak commanded joint speed dropped from about twelve times the manufacturer's stated limit to exactly one times it. The old code had never read the limits file at all.

This is worth stating plainly because it was not a tuning problem. Sweeping the old method's step size across a wide range does not make the straying go away; it settles at about 68 mm and stays there. The fault was the independence of the limits, not their size.

4.6 What the simulation does not tell us

Every joint on the real arm stops slightly short of where it was sent, on the side it approached from. Measured across 252 recorded joint movements, 211 of them land between 0.25° and 0.30° short, with the direction following the direction of travel. That is a bound rather than a spread, which means it is a systematic offset and not noise.

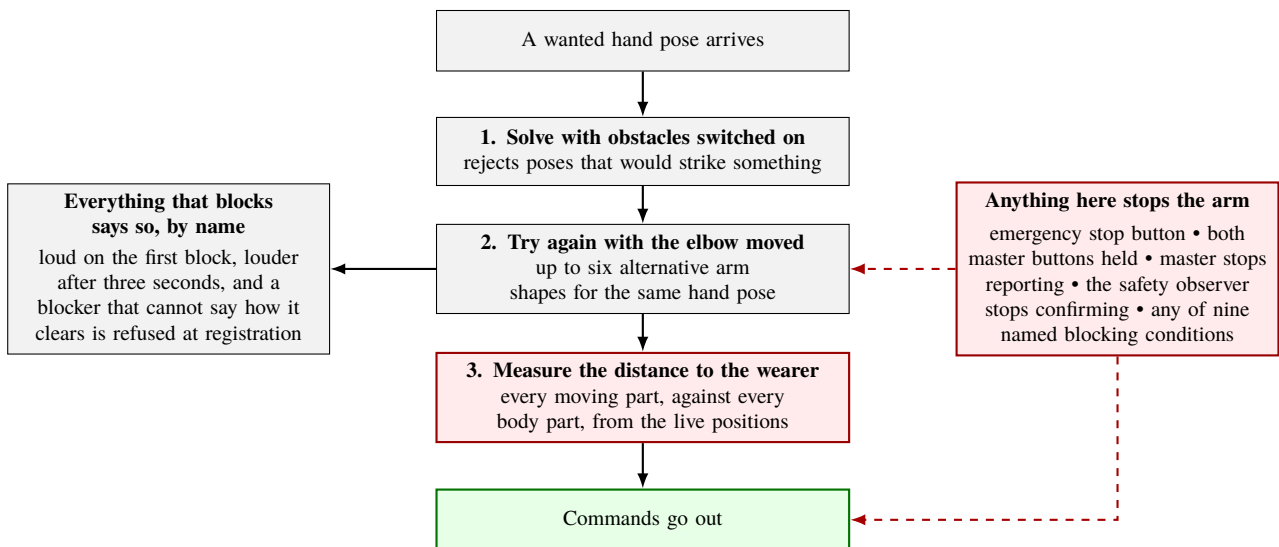
At the hand, that adds up to about 7.2 mm on a 100 mm move. A single correction per arm, which simply overshoots each joint slightly in the direction it is already going, removes about three quarters of it. The correction is implemented and switched off, because it has not yet been tried on the physical arm and switching it on untested would change every recorded figure at once.

Chapter 5

Keeping the wearer safe

The wearer cannot see what the driver intends, cannot feel the controls, and has a moving machine within reach of their head. This chapter describes what sits between the two, and, more usefully, the ways that layer can appear to work while doing nothing.

5.1 Three stages and a floor



Stage 3 is measured from the shapes of the arm and the body directly. It does not use the simulator's own collision settings, because those deliberately ignore the parts of the arm closest to the wearer's shoulders and would report a clear pose with the tube inside the person.

Figure 5.1: What every command passes through. The first two stages reject or reshape a pose. The third measures the distance from the arm to the person and holds if it is too small. Anything on the right can stop the arm outright, and everything that stops it has to say so by name.

The first stage asks the solver for joint angles with obstacle checking switched on, so a pose that would strike the table or the frame comes back refused. The second stage handles the common case where the hand pose is fine but the arm has chosen an awkward shape to reach it: the solver is asked again from up to six different starting guesses, which swings the elbow around without moving the hand. The third stage is the one that matters, and it is described next.

5.1.1 Why the simulator's own collision check is not enough

Robot software of this kind ships with a file listing which pairs of parts are allowed to touch. Pairs that are permanently adjacent, such as the arm's base and the bracket it is bolted to, are excluded, otherwise every pose would be reported as a collision.

On a bench robot that is harmless. On this one it is not, because the parts nearest the wearer are exactly the parts that had to be excluded. Forty-four such pairs are excluded here, and every one of them is a piece of arm that swings past a shoulder, a neck or a chest. A pose the standard check calls valid can have the upper arm tube inside the person.

The clearance check therefore does not use that file. It takes the shapes of the moving arm and the shapes of the body, from their live positions, and computes the shortest distance between them. If that distance falls below 150 mm, the command is held, nothing is published, and the log names which body part was too close.

Two facts about that number are worth stating. The mounting bracket itself sits 161 mm from the wearer's torso and no joint moves it, so the margin has only 11 mm of headroom before the arm has done anything at all. And a clearance reading of exactly 161 mm is a measurement of the bracket rather than of the arm, which is why so many figures in this project sit at that value.

5.2 Where the body is

Until late in the project the body in the model was a mannequin, and the clearance check measured against that. A camera was then added that watches the wearer and estimates where their limbs actually are, using an off-the-shelf body-tracking model [15]. Merging a measured body with an assumed one is where this could go wrong, so the rule is deliberately one-sided.

The camera may only ever make the wearer *bigger*.

Part by part, whichever of the measured shape and the assumed shape is *closer* to the robot is the one the check uses. A tracking result placing the body further away than assumed changes nothing at all. This was tested by injecting a body 0.1 m, 0.3 m, 1 m and 5 m further away in turn; none of them moved the enforced margin. Six deliberate faults were then fed through the running system, including a dark frame, an empty frame and two people in shot, and each refusal was required to name the fault it had caught. That last requirement is what exposed a check that had never actually run.

The honest limitation is on the other side. If the wearer raises their arms and the tracking then drops out, the system falls back to the assumed posture, which has the arms down. Measured through the guard's own geometry, arms held out to the sides leave 93 mm of clearance at the home pose, where the fallback believes there is 317 mm. The current answer to that is an instruction to the wearer, not a control system, and [Chapter 10](#) treats it as the open safety problem it is.

5.3 Stopping

Three things stop the arms, and they are independent by design: a button, both master buttons held together, and the master going quiet.

The third is the one that taught the most. A dead-man switch of this kind watches for the driver's input to stop arriving. The implementation watched for messages to stop arriving, which sounds like the same thing and is not.

When the master's sensors degrade, the software that reads them does not go silent; it substitutes the last good reading and carries on publishing at the same rate. The topic never went quiet, so the dead-man could never fire on the failure that actually happens. It fired reliably when the cable was pulled out, which is why it passed every test for months.

The fix was to stamp each frame with the age of the data it was built from rather than the time it was sent, and to key the dead-man on that. Measured afterwards: a master publishing steadily but with frozen data trips in 0.94 to 0.99 s; a genuinely silent master trips in 0.97 s; and 243 healthy frames produce no false trip.

A second fault in the same mechanism is worth recording because it was invisible in the same way. The stop routine tried to notify the arm driver first, using a call that waits up to two seconds for a service that does not exist when running in simulation. It did that twice. So every emergency stop blocked for four seconds before parking the arms, before logging anything, and before the rest of the system could be told. The detection was always on time. The response was not, and the log line that would have revealed it was itself stuck behind the delay. Parking now happens first and the driver is notified afterwards.

5.4 Making blocking visible

Four separate mechanisms in this system can stop all motion, and each of them is usually right to. The problem was never the stopping. It was that a stopped arm and a working arm looked identical from outside.

Everything that can stop motion now registers with one component that enforces three rules. It must announce itself by name the first time it blocks and every five seconds after. Blocking for more than three seconds is escalated, because sustained blocking is never normal. And a blocker has to state, at the moment it registers, how it will clear; one that cannot is rejected outright, because a condition with no exit is a latch.

That last rule came out of an incident which repeated five times, the fifth inside the code written to catch the first four. The pattern is always the same: the code that clears a condition sits after the code that sets it, so if setting the condition also stops that routine running, the condition can never be cleared. Rather than police each instance, any blocker that stops being re-asserted for two seconds now expires, and an expiry is treated as a third state rather than as an all-clear. Nobody asserting a condition is not evidence that the condition went away; it is evidence that whatever was watching has stopped.

5.5 Testing by breaking things

The safety layer is validated by injecting faults rather than by reading the code. Fourteen faults are injected in turn, covering sensor dropout, the master disconnecting, an emergency stop during motion, the object-tracking losing sight of its target, a logging failure, a clock going backwards, an attempt to write identifying data, the arm's control session being lost and the network to an arm going down. All fourteen are handled.

Getting there was less tidy than that sentence suggests. Four separate bugs in the injection harness each produced a clean table of results while testing nothing at all. A fixed delay was not reliably landing inside a trial. The publisher had not finished being discovered before the fault was sent. State left over from the previous run satisfied the trigger condition for the next one. And the abort signal was a one-shot event, so a fault raised in the gap between two trials was seen by neither.

The transferable lesson is not about this system. Any fault-injection harness is itself an instrument, and its verdicts should not be believed until it has been shown capable of reporting a failure.

Table 5.1: The fourteen injected faults and the response required of each. All are currently handled.

Injected fault	Required response
A sensor channel drops out	Channel disabled automatically, loss of capability announced, trial marked invalid
The master is unplugged	Dead-man trips within one second, arms park
Emergency stop during motion	Arms park, condition latches, clears only on an explicit reset
Object tracking latches the wrong target	Estimate switches within 0.1 s
Two targets equally likely	No assistance offered; a tie is not a decision
The tracked object disappears	Estimate cleared rather than held
Nothing is detected at all	Assistance yields to the driver
The log cannot be written	Raises rather than dropping samples
Identifying data is submitted	Refused in code
The clock goes backwards	No interval measured from wall-clock time
The master reconnects	Detected again, next trial gated until data is fresh
The arm's control session is lost	Closed and reopened without a restart
The camera sees nothing before a trial	Trial refused before it starts
The network to one arm drops	<i>Both</i> arms frozen

The last row is a design decision rather than a technical one. One arm live and one arm frozen on a person's back is worse than both frozen, and the two cannot be told apart by looking.

5.6 Two problems the desk arrangement introduces

When the driver holds tracked hand controllers across the room, the headset sits on a shelf and acts as the fixed reference for everything the controllers report. Two failures follow from that and neither announces itself.

The first is that the shelf can be knocked. Nothing breaks: the controllers stay tracked, the update rate stays at 90 Hz, every reported position remains valid, and all of them are now expressed in a frame that has quietly rotated. The system latches the reference at the start and freezes if it moves by more than 20 mm or 2°. The rotation limit is the one that matters, because a small twist barely moves the headset and puts every subsequent command several degrees wrong.

The second is that a controller can be blocked from view. The system had a signal named for exactly this and it was derived from when the last message arrived, not from whether the controller in it was being tracked. Covering a controller with a hand for five seconds produced no reaction at all while the stream ran on normally. When the driver wears the headset this hardly matters, because their hands are in front of its cameras. With the headset on a shelf, being blocked from view is routine.

SPACE RESERVED FOR A SCREENSHOT

The safety view during a run: the wearer drawn with the 150 mm margin shown as a shell around them, the nearest part of the arm highlighted and labelled with which body part it is closest to, and the readout showing the current clearance. Take one shot in a safe pose and one with the arm deliberately brought close, so the difference is visible.

Figure 5.2: The clearance view. It draws the same body the check is enforcing against, so what is displayed and what is enforced cannot drift apart.

Chapter 6

Seeing, and deciding

6.1 The problem with written-down coordinates

For most of this project the robot was told where things were. The blocks sat at particular coordinates because a file said so, and the table was at a particular height because a file said so. Every one of those numbers was checked, repeatedly, along the whole path the arm would take.

None of it survives a different table. Or a different object. Or the same object moved by five centimetres. The number in the file becomes a statement the arm acts on with complete confidence, and there is nothing anywhere in the system to disagree with it.

The parts needed to measure instead already existed and had never been connected to each other. One module could find a flat surface and the things standing on it, and was being used to print a description. Another could hold a map of the space so the planner could avoid things that are not the wearer, and had never been built outside its own self-test, which means no path this project ever planned knew that the table was there. A third existed so the surface height could come from the camera, and nothing called it.

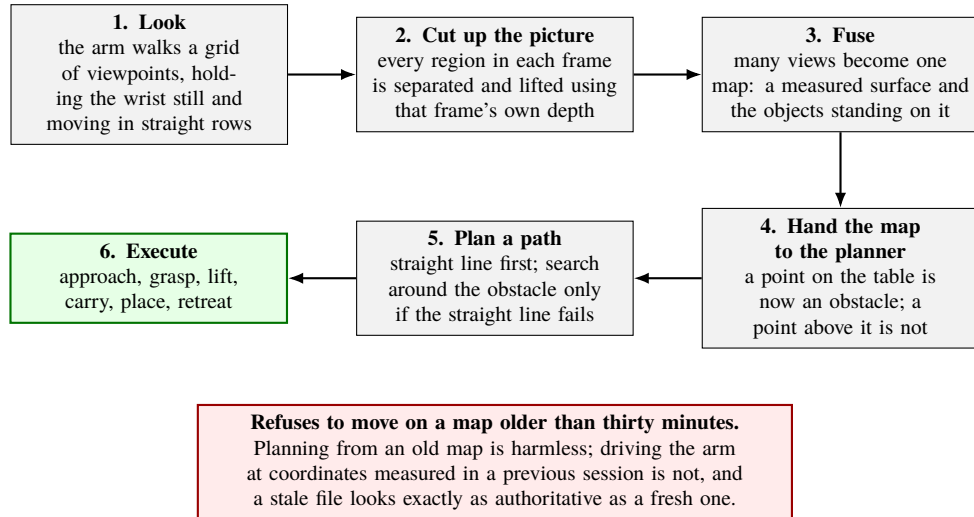
Figure 6.1 is what replaced that.

6.2 Looking like a printer, not like a person

The first version of the sweep pointed the camera at each grid point in turn. That seems sensible and it is wrong. Re-aiming means the wrist is at a different angle in every frame, so any small error in where the camera sits relative to the wrist becomes a *different* error in every frame, and can never cancel out or be measured.

A three-dimensional printer's height probe does not re-aim. It holds one attitude and moves. The sweep does the same: one wrist direction per pass, with the camera positions being the work positions shifted back along a fixed line, so neighbouring views differ by a pure sideways move. Several passes at different angles give the sides of objects, which a single viewpoint cannot see. The full sweep is two heights, three angles, twenty-four positions and two arms.

One more distinction turned out to matter. Being stationary is not the same as having arrived. A joint can sit a small distance from where it was sent and still be moving through that position, and a picture taken then is smeared across two poses while every number downstream looks perfectly healthy. The sweep now waits until no joint has moved by more than a tenth of the arrival tolerance for four hundred milliseconds.



Before this existed, every coordinate the arm used was written in a file by hand. All of it was verified, and none of it survived a different table, a different object, or the same object moved fifty millimetres.

Figure 6.1: Measuring the working area and planning against what was measured. The sweep is deliberately shaped like a printer's bed probe: hold one wrist direction and move in straight lines, rather than re-aiming the camera at every point.

Table 6.1: What the map contains after a full sweep, against the scene it was measuring. Objects are cut out of the picture and lifted through the depth image rather than being clustered from the point cloud.

Quantity	Measured
Surface height	1.2503 m against a true 1.250 m
Surface tilt	0.036°
Objects found	6 of 6
Object centres	every one within 1 mm in both horizontal directions
Object sizes	every dimension within 3 mm
Viewpoints reached	74 of 74 that the arm can reach
Time	13 min 15 s

SPACE RESERVED FOR A SCREENSHOT

The measured map drawn over the real scene: the fitted surface, the six recovered objects with their outlines, and the arm at one of the sweep viewpoints. A second panel showing the same map handed to the planner, with the table drawn as an obstacle, would make the hand-over visible.

Figure 6.2: The result of a sweep. Everything the planner knows about the room comes from this, and it carries the time it was made so that a stale map can be refused rather than trusted.

6.3 Finding a particular object

Detection uses an open-vocabulary detector that takes a phrase rather than a fixed list of classes [5], with a fast segmentation model to get the outline [29] and a plain colour-and-depth filter as a fallback when neither model is loaded. Which of the three produced a given detection is reported with the detection, so a result can never quietly imply that the good one was running.

Two practical points are worth recording. Colour alone does not separate the target from the laboratory: the green block and the teal robots on the far bench differ by two units of hue, which is inside the noise. They are also 2.8 m away while the rig works within 1.6 m, so distance separates them cleanly and colour cannot. And the grasp itself is computed geometrically from the shape of the depth points rather than learned, because a two-fingered gripper closing on a rigid object is a solved problem, a learned model would need graphics memory this machine does not have, and a geometric method can explain why it refused.

That last property earns its place. Every refusal names a cause: no depth data, wider than the gripper's 85 mm opening, too few points to fit a shape.

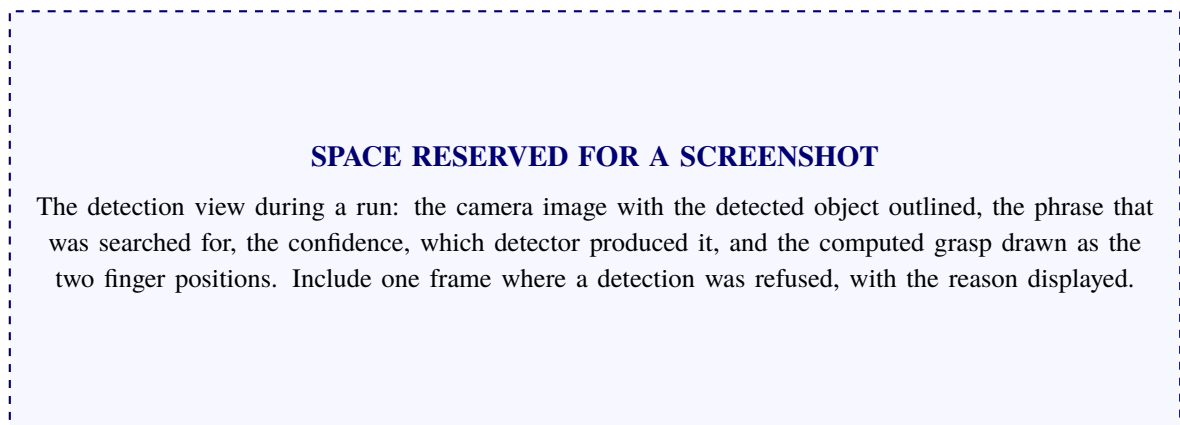


Figure 6.3: What the driver is shown before the arm moves. The refusal case is the one worth including: a system that cannot show why it declined is one an operator will learn to override.

6.4 Planning around things that are not the wearer

For teleoperation there is no planner, and that is correct: the driver's hand is the plan. For the autonomous mode there is a goal, and a planner was written for it.

It searches in joint space rather than in the space the hand moves through, because the thing that has to be avoided is the distance from the whole arm to the wearer, not from the hand [13]. It checks the same clearance the safety layer enforces, rather than the simulator's collision settings, for the reasons given in Chapter 5. It checks every intermediate point along each candidate move at the same spacing the follower uses, because a planner that checks only the ends of its moves has the same defect it exists to remove. And it tries a straight line first, since most moves are fine and planning a clear move turns sixty milliseconds into several seconds.

Why this was needed can be stated in one measurement. A pick whose starting pose and grasping pose are both individually clear, executed as a straight line in joint space, passes within 146 mm of the wearer at step twenty-nine of sixty-four. The path checker catches it, and catching it means the run stops.

6.5 Taking an instruction in English

The third way of driving takes a sentence. The parser is a grammar, not a language model, and the reasons are given in [Chapter 3](#).

The important design decision is what happens when a sentence is ambiguous. Asked to *put the blue block on the blue mat* when there are two blue blocks, the system asks which one rather than choosing. Nearest, leftmost and first-detected are all defensible and all guesses, and this arm is bolted to a person. Asked to *put the blue ones on the blue mat*, it does not ask, because the plural settles it. The difference lives in the words rather than in a rule about how many things happen to match.

Table 6.2: Seventy-five phrasings of the same task, run through the current grammar and through the version in use six days earlier. The two columns differ by the code and by nothing else: the older grammar was checked out of version control and run in its own interpreter.

Outcome	Earlier grammar	Current
Understood correctly	21	34
Asked a clarifying question	14	14
Refused	37	27
Misunderstood	3	0

The bottom row is the one that matters. A refusal wastes the driver's time; a misunderstanding moves the arm. All three misunderstandings in the earlier version were compound instructions of the form *put the blue ones on the blue mat and the green ones on the green mat*, where the grammar planned the first half, dropped the second in silence, moved two objects of four, and reported success.

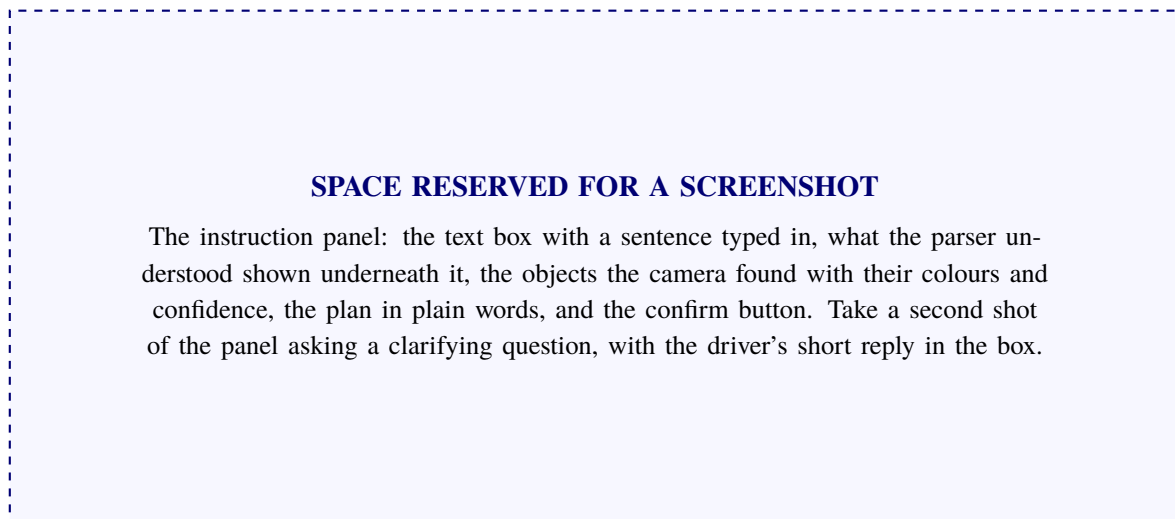


Figure 6.4: The instruction panel. Nothing moves until the driver has seen what was understood and confirmed it, and the sentence is passed to the runner exactly as typed so the record of what was asked cannot drift from what was done.

Chapter 7

The study

This chapter sets out the study the system was built for: what would be asked of the participants, what would be measured, and how the data would be analysed. None of it has been run. Stating it in full is what makes [Chapter 9](#) an empty container rather than a gap.

7.1 The question

Does letting the machine take over part of the work help, and who does it help?

The second half is the part that is usually left out. In a two-person arrangement, an improvement for the driver is not automatically an improvement for the wearer, and the existing evidence suggests it may be the opposite [31]. The study is therefore designed so that both people are measured on every trial, and so that a result can come out mixed.

7.2 Design

Sixteen pairs of participants, each pair doing every available combination of condition and task, three trials each. Everything varies within a pair rather than between pairs, because the differences between two people are larger than the differences the study is looking for.

Condition order is balanced using a scheme that balances not only how often each condition appears in each position, but also which condition follows which. Simple counterbalancing leaves the second kind of imbalance in place, and with a task that people get better at, it matters.

Two roles, not one. Each participant does one role for the whole session. Swapping them mid-session would halve the sample and confound role with fatigue, and the wearer's role involves carrying seventeen kilograms, which nobody should do twice in one visit.

7.3 The tasks

T1, reaching. Six marked targets, three per arm, reached in a given order. No grasping. This is the only task available in all three conditions and is therefore the one that carries the main comparison. Difficulty is indexed in the standard way from the size of the target and the distance to it [7], so trials of different geometry can be pooled.

	T1 reaching six targets, no grasping	T2 pick and place four blocks to matching mats	T3 two-arm carry one object, both grippers	T4 hand to wearer the robot serves the person carrying it
C1 direct teleoperation	3 trials per dyad	not currently possible	not currently possible	3 trials per dyad
C2 assisted teleoperation	3 trials per dyad	not currently possible	not currently possible	3 trials per dyad
C3 spoken or typed goal	3 trials per dyad	3 trials per dyad	not currently possible	3 trials per dyad

Orange cells are not a design choice. Grasping under direct teleoperation is not achievable on this platform, and the two-arm carry needs a grip wider than the hand opens. Both are measured results reported in [Chapter 8](#), and both are stated here so that the empty cells in [Chapter 9](#) are not read as missing data.

Sixteen pairs of participants. Every pair sees every available cell. Condition order is balanced across pairs so that no condition is systematically first or last, and so that each condition follows every other equally often.

Figure 7.1: Three ways of driving, crossed with four tasks, within each pair of participants. Cells marked as not currently possible are measured limits of the platform rather than omissions, and are explained in [Chapter 8](#).

T2, pick and place. Four coloured blocks on a table, two coloured mats, each block to the mat of its own colour. This is the task the autonomous mode performs end to end from a typed sentence.

T3, two-arm carry. One object held by both grippers and moved without dropping or tilting it. Currently not performable, for a reason given in the next chapter: the object presents more width to the gripper than the gripper opens.

T4, handing to the wearer. The robot picks something up and offers it to the person carrying it. It is the only task in which the machine serves the wearer rather than working past them, and it is the one where ownership and trust have something to attach to.

7.4 What is measured

Table 7.1: Everything recorded, by whom it is answered and when.

Measure	Source	When
Completion time	Logged automatically	Every trial
Path efficiency	Logged automatically	Every trial
Success or failure	Logged automatically	Every trial
Number of clutch re-engagements	Logged automatically	Every trial
Closest approach to the wearer	Logged automatically	Every trial
Emergency stops and their cause	Logged automatically	Every trial
Task load index [9]	Both participants	After each block
Physical exertion [3]	Wearer	After each block
Trust in the system	Both participants	After each block
Sense of ownership [14]	Wearer	End of session
Free comment	Both participants	Debrief

Exertion is asked of the wearer between every block rather than once at the end. Fatigue in a task like this rises

steadily, and asking once at the end turns it from something measured into something guessed at.

7.5 How a session runs

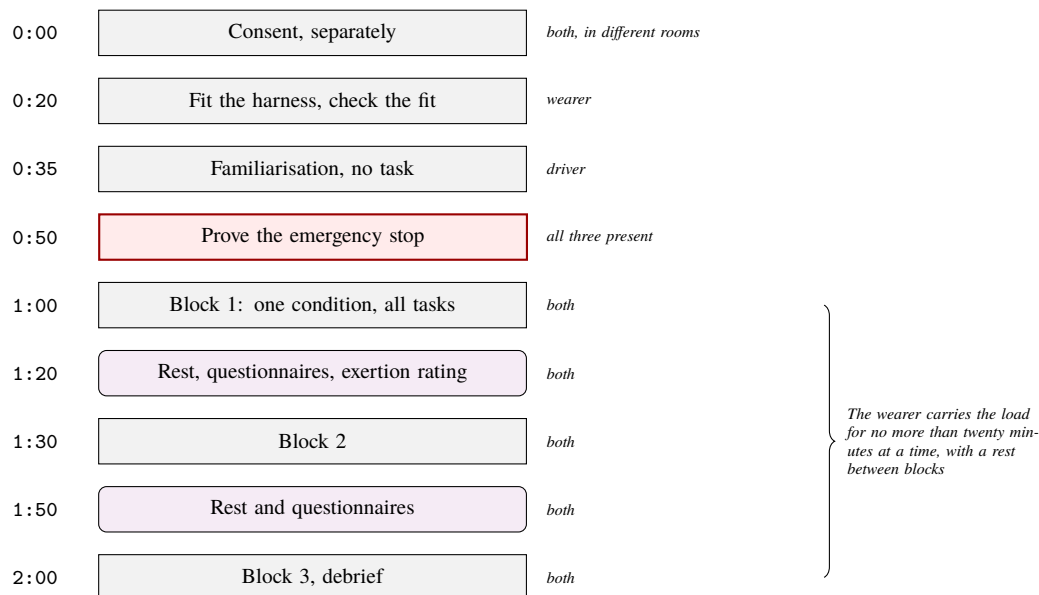


Figure 7.2: One session, about two hours. The red row is not optional: the emergency stop is demonstrated and tested with both participants watching before any task begins, and the wearer is shown how to stop the arms themselves.

Three people are present. The two participants, and a third person whose only job is to hold the stop and watch the wearer rather than the screen. That third role exists because the driver is concentrating on the task and the wearer cannot see behind their own shoulders.

Consent is taken separately and in different rooms. The two participants may know each other, and a wearer who feels watched by the person who brought them is not consenting freely.

7.6 Analysis

For each measure, the comparison is across conditions within a pair. Where the assumptions hold, a repeated-measures analysis of variance; where they do not, its rank-based equivalent. Post-hoc comparisons are corrected for multiple tests, and both the raw and the corrected values are reported side by side so that a reader can see what the correction did.

Effect sizes are reported with confidence intervals, and the intervals rather than the significance are what the discussion will be written around. With sixteen pairs and three trials each, the study is powered to detect a large effect on completion time and is not powered to detect a small one. That is stated in advance rather than discovered afterwards.

Trials that were invalidated by a fault are excluded, and the rule for exclusion is written into the analysis code rather than applied by hand. If more than half a session's trials are invalid, the analysis stops and says so, because a session that fails silently and produces empty averages is indistinguishable from a null result.

7.7 Ethics and safety

The application is written and the study has not been submitted for approval, because a study cannot honestly be submitted while its baseline condition depends on sensor channels that do not work.

The largest single risk is easy to state: a seventeen-kilogram machine mounted behind a seated person, moving under partly automatic control, within reach of their head. Everything else follows from that.

Table 7.2: The main risks to participants and the measures in place. Software measures are the ones described in [Chapter 5](#); the rest are procedural.

Risk	What is done
Contact with the wearer	A 150 mm margin enforced along the whole path, measured geometrically, with the arm holding rather than slowing when it is reached
The wearer cannot stop the arms	A stop within the wearer's own reach, and a third person holding another
Load on the wearer's back	Twenty minutes carrying at a time, seated where possible, exertion rated between blocks, and the session ended on request without explanation
Fatigue confounding the results	Condition order balanced; exertion recorded as a covariate rather than assumed constant
A participant feeling unable to withdraw	Consent taken separately, and the right to stop restated at every break
Identifying data	Refused in code: the routine that writes the session record raises an error if given a name, an email address, a date of birth, an address or a telephone number

Two things are worth saying about the last row. It is enforced in the program rather than in a policy document, because a policy document does not stop a keystroke. And video recording, which would be useful, is not part of the protocol: a video of somebody wearing this rig is not anonymous, and the analysis does not need one.

Chapter 8

Results

Everything in this chapter was measured on the system itself. No person other than the author has used it, and no result here involves a physical arm moving under load. Where a quantity has never been measured, the text says so.

8.1 When the instrument is the fault

This section comes first because it changes how the rest should be read.

Seventeen times in six weeks, a result that looked like a fault in the robot turned out to be a fault in the thing measuring the robot. Not seventeen small discrepancies: seventeen occasions on which a confident, specific and wrong number was produced and would have been believed.

Table 8.1: A selection of the measurement faults found in this project, with the mechanism in each. The pattern is the useful part: none of them announced itself, and every one produced a plausible number.

What was reported	What was actually happening
Sensor noise of exactly zero on all fourteen channels	The recorder writes rows three times faster than the sensors update, so most rows are copies of the one before
Object detection succeeding on 0–4 % of frames	The synthetic test images were flat coloured rectangles; the same detector scores 0.89 on a photograph
A measure of symmetry reading exactly zero	Sixteen failed lookups in a row. Perfect symmetry and an empty loop produce the same number
A scene rendering correctly at 18 % ink coverage	The scene was empty. The wearer alone is 18 % of the frame
Every waypoint of a task unreachable	The check was solving for the wrist pointing where the arm rests, not where the task actually sends it, 43° away
A clearance of 367 mm	The arm was physically touching the mannequin. The check samples the centres of joints and misses the arm between them, and it returns the same number for a clear pose
An arm reported as stalled 360° from its target	Three of the joints turn continuously, so −179 and +181 degrees are the same position. The move had finished
A test suite reporting a failing test called =	The style checker echoes the lines it objects to, and one source file contained the word FAILED inside a string

The working rule that came out of this is short. A surprising result is evidence about the instrument until the instrument has been cleared. In practice that means every analysis program is given an input whose answer is already known and has to return it; a check that cannot be made to fail on a deliberately broken input is not a check; and a result that contradicts an earlier measurement is treated as an instrument problem until the two are reconciled.

The last of those is easy to skip and is the one that pays. Two figures for the same quantity, produced by different programs on different days, disagreed by a factor of five. The instinct is to trust the newer one. Reconciling them found that the older program had been asking a different question for weeks.

8.2 The master arm

Six of fourteen channels currently meet the health criterion, and the distribution of failures is not random: five of the six failures are on one arm, which points at wiring rather than at the sensors.

Replaying recorded traces with the driver's hand held still, and with only the failing channels allowed to move, gives the size of the problem. On the worse arm the failing channels alone command 54.5 mm of hand movement, against a master reach of 272 mm. On the better arm the equivalent figure is 0.014 mm. There is no filter that rescues the first number, because the values are not noisy: they are wrong and they are steady.

The processing chain itself contributes nothing. Feeding it a constant input produces a constant output to within the deadband, which is the control result that lets the 54.5 mm be attributed to the sensors rather than to the software reading them.

8.3 Where the commands go

Vertical and forward-and-back commands map correctly on both arms. Sideways does not, and the reason is geometric rather than a matter of which sign convention was used.

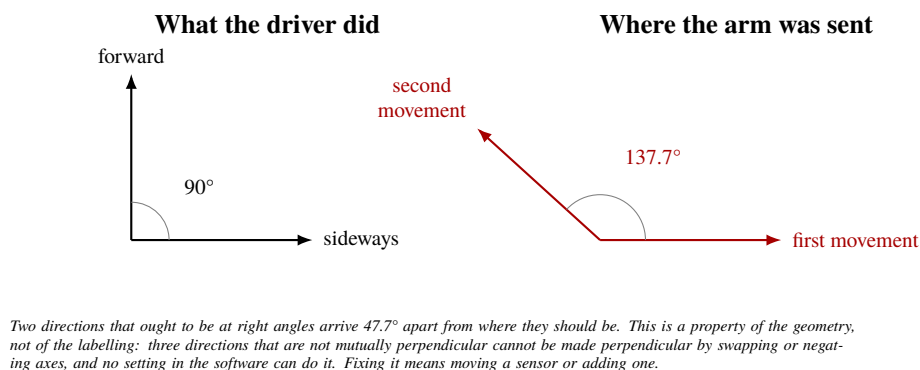


Figure 8.1: Two movements the driver makes at right angles to each other arrive 137.7° apart at the robot. The test is deliberately independent of any axis labelling, which is what makes this a geometric limit rather than a calibration error.

The mechanism is that sideways movement is estimated from a single shoulder sensor whose axis is not aligned with the direction the driver perceives as sideways. Fixing it needs either the sensor moved or a second one added; it does not need software.

8.4 The workspace

This section contains the project's main negative result.

Four separate measurements were made because the first result was surprising enough to distrust. Sweeping the height of the working surface and the distance to it, over 3 360 combinations, with the objects resting on

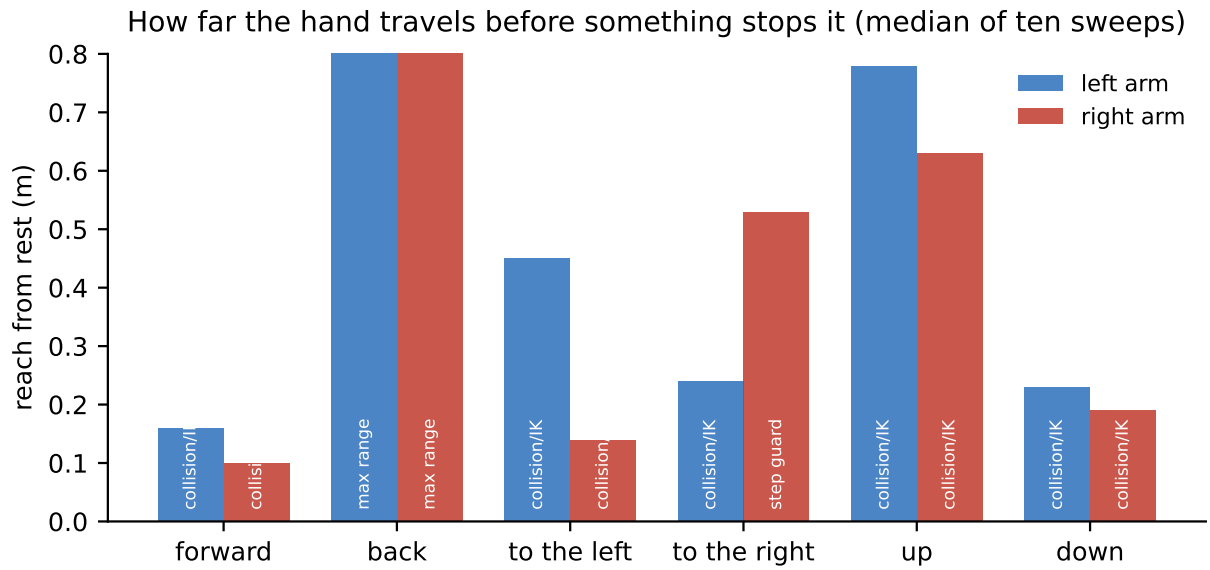


Figure 8.2: Reach from the resting pose in the six principal directions, with the wearer in the model, as the median of ten sweeps. The label inside each bar is what stopped the arm. In sixteen of the twenty-six directions measured, that is either a collision or the arm's inability to fold into the shape required, and almost never the length of the arm. *Max range* means the sweep ran out of travel before the arm did, so those figures are lower bounds.

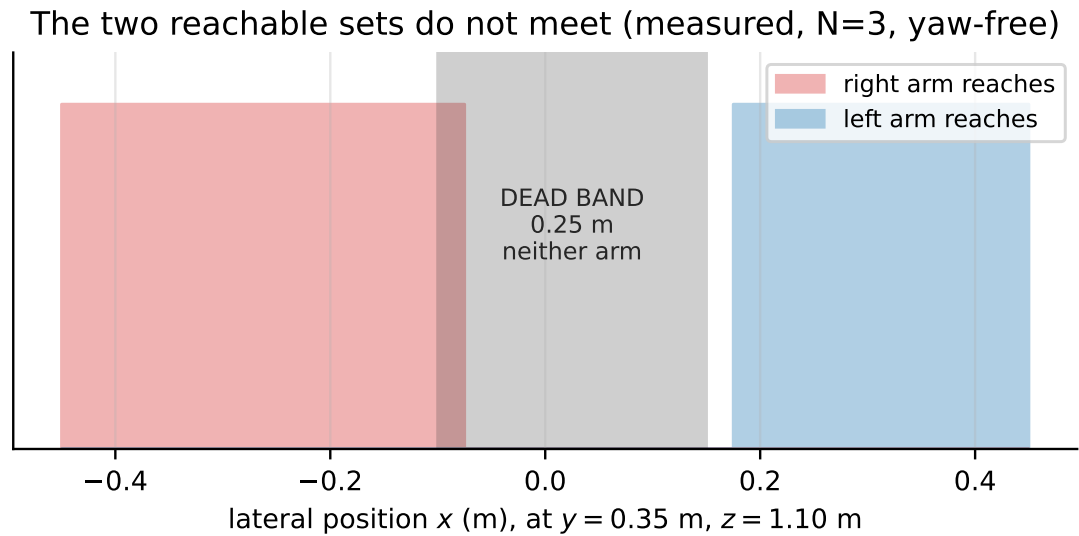


Figure 8.3: Points in front of the wearer that each arm can reach. The two sets do not overlap anywhere, and there is a gap of about a quarter of a metre between them straddling the body's centre line. Measured over a grid of sixty-three positions with the wearer present; with the wearer removed entirely, three shared points appear, and the wearer removes all three.

the surface rather than floating above it, produced no arrangement at all in which an object within 100 mm of the wearer's centre line could be picked up. With the wearer's own arms deleted from the model, the innermost workable column moves in by 25 mm on one side and 75 mm on the other, and the limit is then the torso. Asked for a top-down grasp instead of an approach from the side, the answer is zero positions out of eight hundred and forty, at any surface height.

That is worth restating plainly, because it is the single most consequential thing measured in this project. **The centre of a person's own working space is the one place a pair of shoulder-mounted arms cannot go**, and no change to the control software affects it.

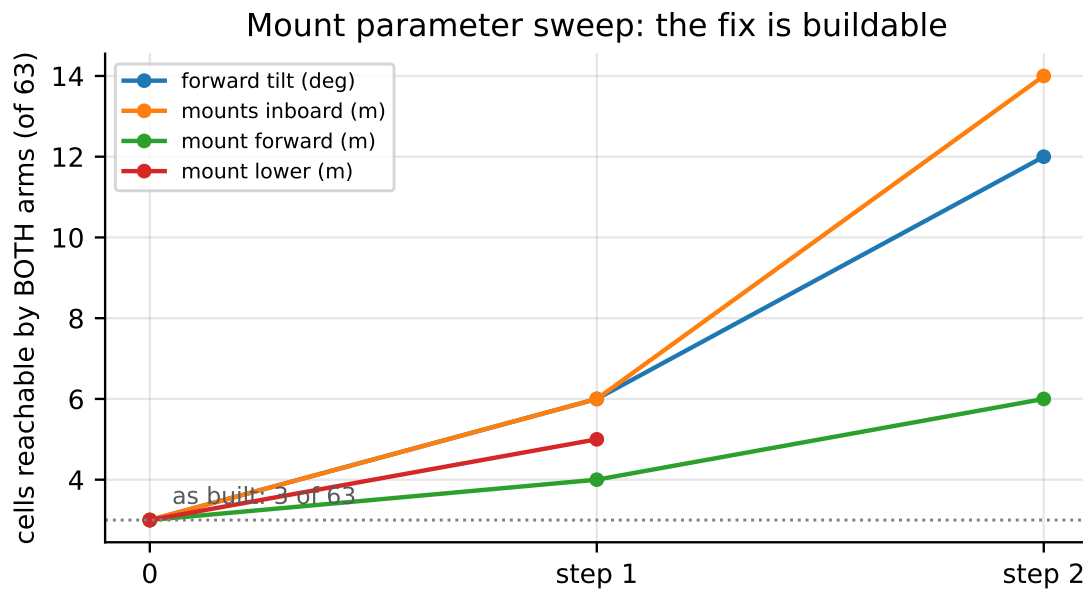


Figure 8.4: Shared reachable positions against each mounting parameter, taken one at a time from the arrangement as built. Moving the mounts inboard and tilting them further forward both help, and they help together. Splaying them outward was measured separately and makes matters worse, which is why it does not appear here. The best buildable combination raises the shared count from three to twenty-two of sixty-three.

The mount change is quantified and has not been applied. Applying it invalidates the resting pose, the approach direction written into every task, and every clearance figure derived from either, so it is a deliberate re-measurement of the whole system rather than an adjustment.

8.5 Grasping, and why one task cannot be done

The gripper opens 85 mm. Measured from the positions of the two finger pads, and taking the width each object presents along the closing direction:

Two of the four tasks are blocked by this table alone, and neither is a control problem. They need a different object or a different gripper.

8.6 Where the hand actually ends up

For a grasp to close on an object the gripper has to arrive within about 30 mm of it. [Figure 8.7](#) is every known contribution to missing that, measured separately.

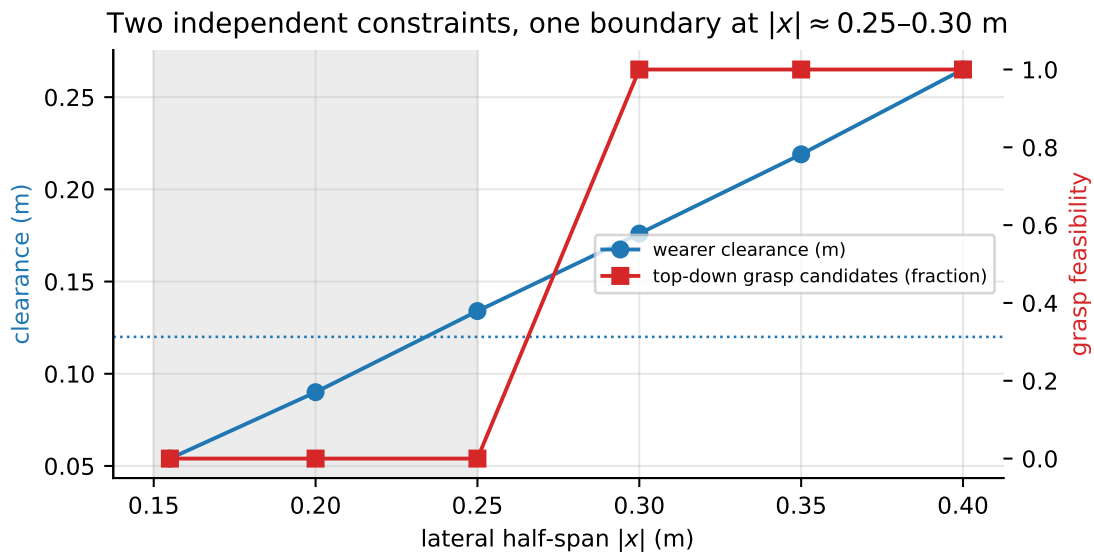


Figure 8.5: Two independent constraints meet at roughly the same place. As the working area is moved outboard, clearance to the wearer rises through the floor at about 0.25 m, and grasping from above becomes possible at about the same distance. The coincidence is not causal, and it means an object placed to satisfy one constraint tends to satisfy the other.

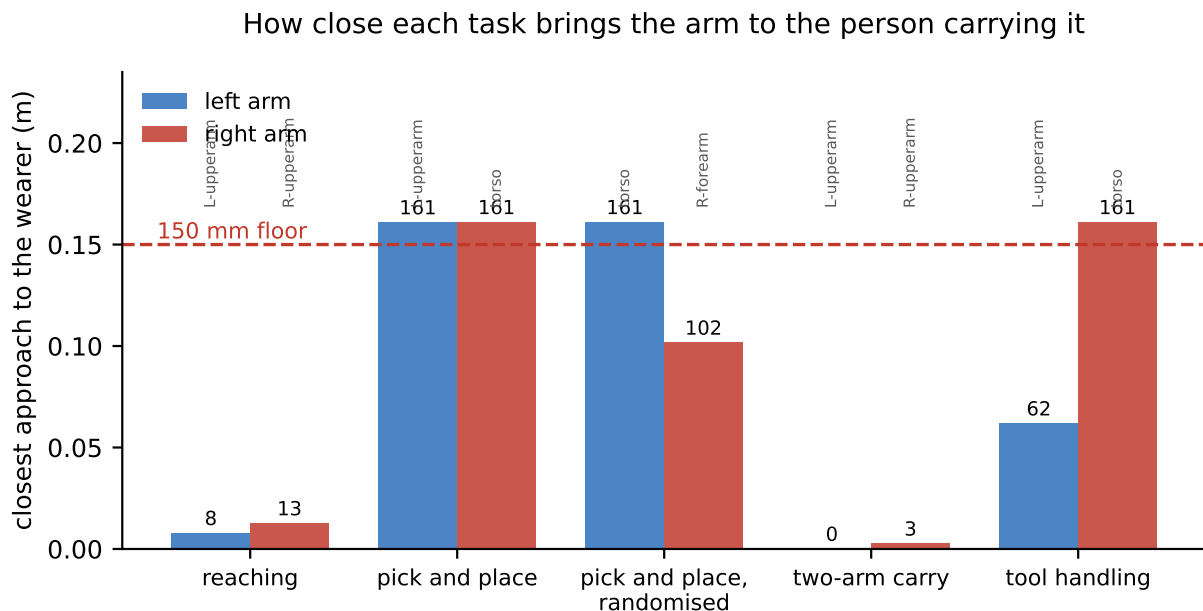


Figure 8.6: Closest approach to the wearer over the whole of each task's path, ten repeats, with the body measured geometrically. The grey label above each bar is the part of the body that was nearest. Only the pick-and-place task stays clear on both arms. 161 mm appears three times because that is the mounting bracket, which no joint moves, so it is the best any pose can do. The breaches are reported rather than quietly fixed: moving a task's coordinates means re-verifying the whole task.

Table 8.2: The width each object presents to the gripper, along the direction the fingers close. The gripper opens 85 mm.

Object	Side approach	From above	Fits?
40 mm block	49.0 mm	40.0 mm	yes, both ways
Carry tray, left grip	83.4 mm	no solution	by 1.6 mm
Carry tray, right grip	113.9 mm	no solution	no
Circuit box	195.0 mm	170.0 mm	no, either way
Multimeter	42.4 mm	30.0 mm	yes

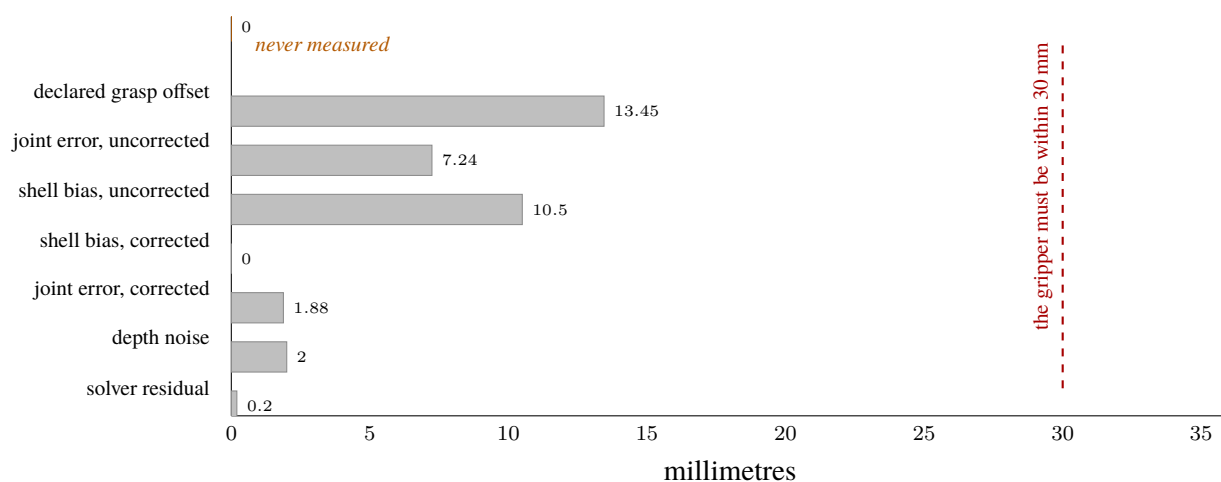


Figure 8.7: Contributions to positioning error at the moment of grasping. Grey bars are measured. The orange bar is the mounting of the camera relative to the arm, which has never been measured and is now the largest unknown. Adding the measured terms in the worst case gives 33.4 mm against a 30 mm requirement; combined in quadrature they give 18.6 mm.

Two of the three largest terms already have corrections written and switched off, and with both enabled the worst case falls to about 17 mm. Both are switched off for the same reason: neither has been tried on a physical arm, and enabling an untested correction would change every recorded figure at once.

The unmeasured term is the one to go and measure. The camera's position is computed from the model of the arm, which is exact given the model, and nobody has ever checked the model against the physical camera module. A first attempt at this on hardware suggests the error is angular rather than a simple offset: the table cannot move, yet the direction of the table surface measured from four different arm positions varied by 8.45° , which is 34 mm of position error at typical working distance. No open-loop grasp planned through that can land inside the gate, and none did.

8.7 The safety layer

All fourteen injected faults are handled. The two timing figures worth quoting: a master that has frozen but is still publishing is caught in 0.94 to 0.99 s, and the emergency stop now parks the arms immediately rather than four seconds later.

8.8 How fast the wearer would have to move

This is the least comfortable result in the thesis and it is not a bug.

The clearance floor is a distance. The guard's knowledge of where the wearer is arrives with a delay of between 62 ms and 562 ms, the spread being a rate limit rather than the tracking itself. Multiplying that delay by plausible limb speeds gives how far a limb travels before the guard hears about it.

Table 8.3: How far a limb moves before the guard knows, against a 150 mm floor. Limb speeds are taken from the literature and are labelled as assumptions, not measurements.

Movement	Delay	Travel	Against the floor
A deliberate reach	62 ms	50 to 93 mm	62 % of it
A deliberate reach	562 ms	450 to 843 mm	exceeds it
A quick reach	62 ms	93 to 155 mm	exceeds it
A quick reach	562 ms	843 to 1405 mm	exceeds it
A flinch	62 ms	155 to 248 mm	exceeds it
A flinch	562 ms	1405 to 2248 mm	exceeds it

In five of the six cases the limb travels further than the entire margin before the system knows anything has changed. The floor is a distance, not a reaction time, and nothing in this system currently reacts to a person moving quickly. [Chapter 10](#) sets out the three things that would help and what each would cost.

8.9 Verification

None of the figures above would be worth quoting without something behind them that fails when the system changes. [Table 8.4](#) is what that amounts to at the date of writing. The two counts worth singling out are the 257 checks made on the operating window, which press every control and then check what the press did, and the fourteen injected faults, because those two are the only checks in the list that exercise the system the way a person would.

Table 8.4: What is checked automatically, as of the date of writing.

Measure	Count
Automated tests passing	1 322
Test files	114
Checks made on the operating window	257
Faults injected and handled	14 of 14
Recorded task demonstrations	17 cells, 8 camera angles each
Grasp success in those recordings	100 %
Analysis and verification programs	233
Stored measurement records	195

8.10 What has not been measured

The list is short and every item is a consequence of the same thing, which is that no experiment has been run on a moving physical arm carrying a load.

Detection accuracy at working distance on the real cameras. The delay from the driver's hand moving to the arm moving, end to end. The behaviour of the control session when the arm's power is interrupted. Whether the sideways mapping problem persists once the master is rewired. How the joint error varies with speed, which was reported as measured and was not: the three speeds turned out to be three repetitions of one condition, because the speed limit was read once when the program started. And speech recognition against a real speaker, which cannot be measured on this machine at all, because it has no audio input device. Every voice figure in this project came from synthesised speech and is a measure of the grammar rather than of the recogniser.

Chapter 9

Results from participants

No participant data has been collected.

This chapter is deliberately complete in structure and empty of results. Every table and figure that the study in [Chapter 7](#) would produce is set out below, with the analysis that fills it named in the caption, so that the chapter can be filled in the order the data arrives rather than rebuilt. [Chapter 10](#) states what has to be resolved before the sessions can run; the first item is two soldered joints.

9.1 Participants

Table 9.1: Who took part, reported separately by role because the two roles are not equivalent: only the wearer carries the load. No identifying field is recorded.

	Drivers	Wearers	Pairs
Number	–	–	–
Age, median and range	–	–	–
Previous experience of teleoperation	–	–	–
Handedness	–	–	–
Sessions completed in full	–	–	–
Sessions ended early, by reason	–	–	–

9.2 Task performance

Table 9.2: Success rate and completion time by task and condition. Cells marked *n/a* are the combinations the platform cannot perform, explained in [Chapter 8](#).

Task	Success rate		Time (s), median	
	Direct	Assisted	Direct	Assisted
T1 reaching	–	–	–	–
T2 pick and place	n/a	n/a	n/a	n/a
T3 two-arm carry	n/a	n/a	n/a	n/a
T4 handing over	–	–	–	–

SPACE RESERVED FOR A SCREENSHOT

Completion time by task and condition. Four panels, one per task, each with one box per condition. The pick-and-place and two-arm carry panels carry fewer conditions than the reaching panel, and the missing conditions should be drawn as an explicit gap with a note, not omitted, so that a reader does not read them as data that was lost.

Figure 9.1: Completion time. To be produced by the autonomy level analysis. Report the omnibus test and the corrected pairwise comparisons, with raw and corrected values side by side.

SPACE RESERVED FOR A SCREENSHOT

Path efficiency and clutch use. Two panels: the ratio of the hand's travelled distance to the straight-line distance, and the number of times the driver disengaged and re-engaged, both by condition.

Figure 9.2: Path efficiency and re-engagement count. These two together say whether assistance made the movement more direct or merely slower.

9.3 Workload and exertion

Table 9.3: Task load index, by role and condition. Reported for both participants because a reduction for one is not a reduction for the other, and the comparison between the two columns is the point of the study.

Subscale	Driver			Wearer		
	C1	C2	C3	C1	C2	C3
Mental demand	–	–	–	–	–	–
Physical demand	–	–	–	–	–	–
Temporal demand	–	–	–	–	–	–
Performance	–	–	–	–	–	–
Effort	–	–	–	–	–	–
Frustration	–	–	–	–	–	–
Overall	–	–	–	–	–	–

SPACE RESERVED FOR A SCREENSHOT

Exertion rating from the wearer, plotted against block number rather than against condition, with the condition marked on each point. If exertion rises monotonically with block regardless of condition, it is fatigue and must be treated as a covariate rather than as an effect.

Figure 9.3: Physical exertion reported by the wearer between blocks.

9.4 Trust and ownership

Table 9.4: Trust, by role and condition, and the wearer’s sense of ownership at the end of the session. The prediction from the existing literature is that trust falls as the machine takes over more, and this study is designed to be able to reproduce or contradict that.

	C1 direct	C2 assisted	C3 autonomous
Trust, driver	–	–	–
Trust, wearer	–	–	–
Ownership, wearer	–	–	–
Agency, wearer	–	–	–

Table 9.5: Everything the safety layer did during the sessions. These are recorded automatically and should be reported whether or not anything happened, because a table of zeros is a result.

Event	C1	C2	C3
Emergency stops, by whom	–	–	–
Clearance holds	–	–	–
Closest approach recorded, minimum	–	–	–
Trials invalidated by a fault	–	–	–
Sessions stopped at a participant's request	–	–	–

9.5 Safety events

9.6 What participants said

To be filled from the debrief. Free comments to be reported by role and grouped by theme, with the number of pairs raising each. Comments from the wearer should be reported at least as fully as comments from the driver: the wearer is the participant with the least control over what happened and the most direct experience of it.

Chapter 10

Discussion

10.1 Designing for sensor loss was worth it, and not for the reason expected

The premise was that a body-worn control arm routing analogue signals through turning joints will lose channels, so losing them has to be survivable. That premise held. Seven of fourteen channels currently fail the health test, and the failures are progressive rather than sudden.

What the architecture actually bought is narrower than it first appears. It did not keep the system working: with the channels in their present state, direct teleoperation is not usable, and no amount of graceful degradation changes that. What it bought was the ability to say so precisely. The system names which channels it no longer believes, states what capability that costs, and refuses rather than inventing a position when too little is left. The alternative, which is what the system did before, was to keep commanding the arm from readings that were confidently wrong.

There is an uncomfortable corollary. A system that degrades gracefully also degrades quietly. Six of the fourteen channels failed over a period during which nothing stopped working in a way anybody noticed, and the failure was found by measuring rather than by using. Any design like this needs the health report to be visible in the ordinary course of a session, not available on request, which is why the channel state now appears in the operating window rather than in a log.

10.2 The reduced-control premise, assessed

The second premise was that commanding fourteen joints is a usability problem, so the system should take position from a few well-behaved sensors and leave the rest to the machine or to nobody.

Taking position from a small number of channels worked. Splitting it into height, direction and distance, and taking each from whichever sensor sees it most directly, is the reason a system with half its sensors failing still produces a usable command on one arm.

Pinning the wrist did not work as well as expected, and the cost was not measured until late. Fixing the wrist orientation on a seven-jointed arm uses up the spare joint, and the spare joint is precisely the one that swings the elbow away from the wearer while the hand stays put. Measured at the point where the work actually happens, pinning the wrist reduced the average distance the arm could travel from 0.30 m to 0.065 m, a factor of seven. Allowing the wrist to vary within a 15° cone recovers most of it and costs nothing measurable in accuracy. This should have been measured a month earlier than it was.

10.3 The workspace result, and what it costs the study

The two arms cannot both reach any point in front of the wearer. That result was checked four different ways precisely because it invalidates part of the study, and it survived all four.

The consequences are concrete. Any task requiring the two hands to meet is impossible, which removes the passing of an object from one arm to the other. Anything requiring both arms on the same object requires the object to be wide enough to span the dead band, which is where the two-arm carry task ran into the gripper's opening. And the innermost column of the working area sits 325 mm from the body's centre line, so a task described as happening in front of the wearer actually happens off to one side.

It is worth being clear about what kind of result this is. It is not a limitation of the control system, and no improvement to the software affects it. It is a consequence of where two arms of this length are bolted to a person of this size. The measured mount change would improve it substantially and has not been applied, because applying it means re-deriving the resting pose, the approach direction and every clearance figure in this thesis. That is a session's work, not an afternoon's, and it should be done as one deliberate pass with everything re-measured afterwards.

10.4 The safety layer is honest about the wrong thing

The safety layer is thorough about distance and silent about time.

Fourteen faults are injected and handled, the margin is measured geometrically against the parts of the arm that actually pass the wearer, and every mechanism that can stop the arm announces itself by name. All of that is sound and most of it was built after a specific failure showed why it was needed.

What none of it addresses is that the margin is a distance and the wearer can move. In five of six modelled limb movements, the limb travels further than the entire 150 mm margin before the guard hears about it. The system is protecting a stationary person from a moving robot, and the arrangement it is deployed in has a moving person as well.

Three things would help and none is done. Inflating the margin by the delay times an assumed speed would at least make the number mean something, and it is a one-parameter change. Cutting the delay is mostly achievable, because most of the 562 ms is a rate limit rather than the tracking itself. And slowing the arm as it approaches the wearer would couple two quantities the system already has and currently does not connect. The first is the one to do first, because it changes what the margin claims rather than merely improving it.

10.5 Measurement was most of the work

The single most striking number in this project is not about the robot. It is that a surprising result turned out to be a fault in the measurement seventeen times.

That has three consequences for how this thesis should be read. Any figure here that has not been produced by a program with a known-answer test behind it should be treated as provisional, and the text says which those are. Any figure that was produced once should be treated as provisional too, because the solver used here restarts from a random guess and a single call is a coin flip. And the absence of a measurement is reported rather than filled in, which is why several sentences in [Chapter 8](#) end in a statement that something has never been measured.

The habit is not free. Clearing an instrument before believing it roughly doubles the time a measurement takes.

Set against that, several of the instrument faults found here would have propagated into published numbers, and two of them into the safety case.

10.6 Limitations

Nothing has run on a moving arm under load. The simulator used here returns zero for every force and torque, so anything derived from a force reading is undefined on this platform rather than merely unmeasured. That is why there is no force feedback result in this thesis and no impedance criterion in the evaluation.

The delay figures are lower bounds. They are measured within one machine, over shared memory. The physical arm adds a network round trip per command, and that has not been characterised.

The mapping results depend on a broken master. The sideways mapping problem is reported as geometric, and the argument for that is a rotation-invariant measurement rather than an inference from the failing channels. But the accuracy figures for the working channels come from an arm whose other half does not work, and they should be re-measured after repair.

One person built and measured all of it. Every design decision was taken by the person who then measured its consequences. The instrument-checking habit was adopted partly as a defence against that, and it is not a complete one.

10.7 What stands between this and the study

In the order it should be done.

1. **Repair the master arm's wiring.** Two of the fourteen channels are dead and six are incoherent, and the failures cluster on one arm. Until that is fixed there is no direct teleoperation baseline, and without a baseline there is no comparison to make. This is a soldering job.
2. **Measure where the wrist camera actually is.** It is the largest unknown in the positioning budget and the reason no grasp has landed from an open-loop plan. It needs the hardware and a calibration target.
3. **Make the clearance check measure the whole arm.** At present it samples the centres of joints, so it can return a comfortable number for a pose in which the arm is touching the mannequin. This is the outstanding safety defect and it should be fixed before anybody wears the rig.
4. **Put the delay into the margin.** One parameter, and it turns a distance that means little into one that means something.
5. **Apply the mount change and re-measure everything.** A day's work and a large improvement in what the arms can reach.
6. **Then submit for ethical approval,** with a baseline that works and a safety case whose numbers have been measured on the arm rather than in simulation.

Chapter 11

Conclusion

11.1 What was done

A complete system was built for driving two worn robot arms and keeping the person carrying them safe. It has three input routes: a sensed mannequin arm, a pair of hand controllers used as motion trackers, and ordinary English typed or spoken. It has a safety layer built on measured distance to the wearer rather than on collision flags, tested by breaking things rather than by reading code. It has a perception system that measures the working area instead of being told about it, and a planner that knows the table is there. All of it is reachable from one window, and all of it is checked by 1 322 automated tests and by a program that presses every button in that window.

11.2 Against the aims

Table 11.1: The six aims from [Chapter 1](#), and what became of each.

Aim	Outcome
Build and characterise the master arm	Done. Six of fourteen channels meet the health criterion, and the failing ones are quantified rather than described
A mapping that degrades predictably	Done, and measured: one step down the ladder costs 63.8 mm of accuracy, and the bottom rung is a refusal
A safety layer suited to a worn machine	Built and validated by injection, 14 of 14. One defect remains open and is named in Chapter 10
Characterise the reachable workspace	Done, and the result is negative: the two arms share no reachable point in front of the wearer
Add two further ways of driving	Done. Both reach the arms through the same safety path, verified by test
Identify what must be fixed before a study	Done, in priority order. The first item is a soldering repair

11.3 What was not done

No participant has used the system, and no result has been validated on a moving physical arm under load. The two-person study is designed, its protocol is written, its analysis is coded and its ethics application is drafted, and none of it has been submitted, because submitting a study whose baseline condition depends on broken sensors would be dishonest.

The two-arm tasks in the study cannot currently be performed, for two separate measured reasons: the arms share no workspace, and the object presents more width than the gripper opens. Both are reported rather than worked around.

11.4 What should happen next

The ordering in [Chapter 10](#) matters more than any individual item. Repair the master. Measure where the camera is. Fix the clearance check so it measures the whole arm rather than the centres of its joints. Put the perception delay into the safety margin so that the margin means what it claims. Then move the mounts and re-measure. Only then run people.

11.5 Closing

The most useful thing this project produced is probably not the system. It is the discipline of not believing a surprising number until the instrument that produced it has been cleared, and the record of seventeen occasions on which that discipline was the difference between a finding and a mistake. The robot will be rebuilt. The habit transfers.

Chapter A

Success criteria and how they were judged

Each criterion below carries a threshold and the reason that threshold was chosen. A criterion with a number attached but no reason for the number is not a criterion.

Table A.1: The nine criteria this work is judged against.

	Criterion	Threshold, and why	Outcome
1	Angle conversion is exact	Converting an angle from the manufacturer's convention to the software's and back must return the original. Any deviation is a wrapping mistake, and the failure mode is nearly a full turn of a joint beside a person	Met
2	Commands go where intended	Within 30° per direction. Beyond that the driver has to consciously correct rather than move naturally	Partly met: up, down, forward and back pass; sideways does not
3	The arm follows the command in simulation	Within 5 mm. Below that the residual is the deadband rather than the controller	Met, 0.7 to 4.7 mm
4	Distance to the wearer	At least 150 mm at all times	Not met on three of the four tasks, reported per task
5	Every injected fault is caught	All of them. A mechanism that catches thirteen of fourteen is not a safety mechanism	Met, 14 of 14
6	Failed channels inject nothing	Below 1 mm of commanded movement from a still hand, and a refusal when nothing usable remains	Partly met: one arm passes at 0.014 mm, the other fails at 54.5 mm
7	Every declared position is reachable	Ten repeats out of ten, along the whole path. A single successful attempt is a coin flip	Met, 17 of 17
8	End-to-end delay	Under 200 ms	Not measured. The highest-value remaining gap
9	Grasps are verifiable in the recordings	The fingers must open, close on the object and reopen, and the object must move with the hand	Met, all recorded grasps

Two criteria are deliberately absent. There is no force or stiffness criterion, because the simulator returns exactly zero for every force reading, so any quantity derived from one is undefined on this platform rather than merely unmeasured. And there is no wrist-tracking criterion, because the wrist orientation is pinned and the channels that would command it are the dead ones.

Chapter B

Equipment

Table B.1: Hardware and software used.

Item	Detail
Robot arms	Two Kinova Gen3, seven joints each, 8.2 kg apiece, 0.902 m reach [12]
Grippers	Robotiq two-finger, 85 mm opening
Wrist cameras	Colour and depth, one per arm
Frame	Backpack harness, arms mounted behind the shoulders with a 60° forward tilt
Master arm	Scale model of the same joint sequence, rotary sensor per joint, inertial sensor at the wrist
Microcontroller	Teensy 4.1, reading all channels at about 50 Hz [22]
Hand controllers	Two tracked controllers, used with the headset stationary as a reference
Computer	Laptop, Linux, graphics card with 4 GB
Robot software	ROS 2 [16], inverse kinematics by TRAC-IK [1]
Motion smoothing	Ruckig [2]
Detection	Open-vocabulary detector [5] with fast segmentation [29]
Body tracking	MediaPipe Pose [15]

Chapter C

Where the evidence lives

Every figure quoted in this thesis is produced by a named program and stored as a file, so it can be recomputed rather than taken on trust. The table below maps the main claims onto the programs that produce them.

Table C.1: Claims and the programs that produce them.

Claim	Produced by
Channel health verdicts	trajectory_capture/channel_report.py
What a lost channel costs	scripts/measure_capability_ladder.py
Reach by direction	scripts/measure_workspace.py
The two arms share no workspace	scripts/sweep_mount_overlap.py
What a mount change would buy	scripts/sweep_mount_geometry.py
Distance to the wearer, per task	scripts/measure_clearance_region.py
Positioning error budget	scripts/measure_control_budget.py
Motion smoothing	scripts/measure_teleop_motion.py
Fault injection	srl_teleop/fault_injector.py
Instruction handling	scripts/sweep_t1_instructions.py
Map accuracy	scripts/calibrate_environment.py
Window checks	scripts/verify_gui_buttons.py
Automated tests	scripts/check_tests.py

Chapter D

Participant documents

To be attached before submission: the participant information sheet, the two consent forms, the pre-session screening questions and the debrief script. Two consent forms rather than one, because the two roles carry different risks: only one of them involves carrying seventeen kilograms, and only one of them involves a machine moving beside their head under someone else's control. The drafts exist in the project's research folder and are not reproduced here because they have not been through ethical review.

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